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Finite element modeling of blood flow in arteries

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Abstract

A comprehensive finite element framework to enable the conduct of computational vascular research is described. The software system developed provides an integrated set of tools to solve clinically relevant blood flow problems and test hypotheses regarding hemodynamic (blood flow) factors in vascular adaptation and disease. The validity of the computational method was established by comparing the numerical results to an analytic solution for pulsatile flow as well as to published experimental flow data. The application of the finite element method to qualitatively and quantitatively assess the blood flow field in a number of clinically relevant models is described. © 1998 Elsevier Science S.A.

1. Introduction

The human vascular system is comprised of the heart and blood vessels, and functions in such a way as to supply each organ with the proper amount of blood, which may vary depending on physiologic conditions and organ demands. The arteries supplying blood, must respond to local changes in the end organs while maintaining overall homeostasis of the circulatory system. Under normal conditions, the arteries can adapt to changes in blood flow and blood pressure. However, under certain circumstances, the adaptive and healing processes fail and the arteries are unable to respond to the imposed forces and, in the worst cases, are unable to provide blood to important organs.

In order to understand the normal and pathologic behavior of the human vascular system, detailed knowledge of blood flow and the response of blood vessels is required. Hemodynamic disturbances have been strongly correlated to the localization of atherosclerotic lesions on the vessel walls. Clinically relevant plaque deposits are most common in areas of complex flow in the coronary, carotid, abdominal, and femoral arteries [46]. These complex flow regions often occur due to branching, bifurcations and curvature of the arteries.

In the carotid artery, atherosclerotic lesions localize in the carotid sinus, the enlarged region just beyond the bifurcation, and can restrict cerebral blood flow. Ultimately, carotid bifurcation atherosclerosis can result in plaque disruption and clot formation culminating in stroke. Fig. 1(a) depicts cross-sections of a carotid bifurcation obtained from a cadaver and illustrates the localization of the disease to the outer wall of the carotid sinus region. Section A, proximal to the bifurcation, and section D, distal to the carotid sinus, are devoid of plaque in comparison with sections B and C in the carotid sinus region. In addition, it is noted in sections B and C that the disease within the carotid sinus is localized to the outer wall. Fig. 1(b) depicts a hydrogen bubble visualization of steady flow in a model of the carotid bifurcation. Flow is rapid, laminar and axial along the inner wall of the carotid sinus, an area of high wall shear stress and low particle residence time. Conversely, the outer wall of the carotid sinus is an area of low wall shear stress and high particle residence time. Low shear

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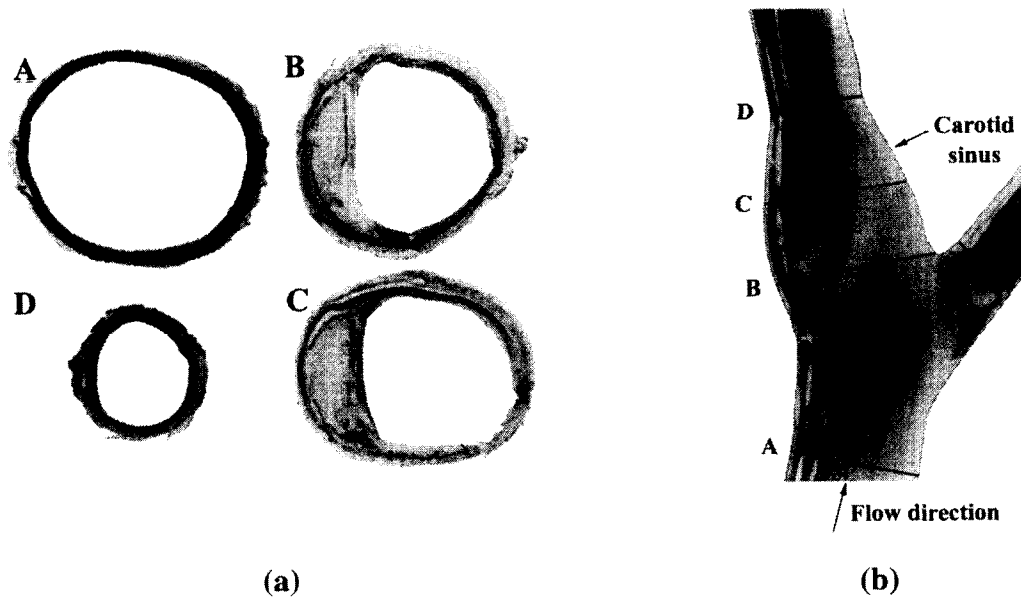


Fig. 1. Localization of plaque within the carotid sinus. (a) Cross-sections from a cadaver; (b) hydrogen bubble visualization of steady flow characteristics. (Adapted from Zarins et al. [43].)

stress and high particle residence time are hypothesized to be ‘hemodynamic risk factors’ for vascular disease [24].

A common surgical procedure is the construction of an end-to-side vascular bypass graft in which the end of a graft is connected to the side of a host vessel. These grafts are frequently employed by surgeons to bypass a diseased or injured segment of an artery to restore blood flow to an organ, e.g. the heart. One of the primary causes of long term graft failure is intimal thickening, a process in which the inner wall of a blood vessel increases in thickness often leading to graft occlusion and failure. Fig. 2 depicts the common sites of intimal thickening for an end-to-side vascular graft [2].

The intimal thickening along the suture line is generally attributed to the normal healing process, but the mismatch between the compliance of the host artery and graft has also been hypothesized to be a factor. The intimal thickening observed at the toe and floor in Fig. 2 is hypothesized to be due, at least in part, to hemodynamic factors.

Fig. 3 shows the results of an experimental flow visualization study in which it was noted that flow recirculation and stagnation regions develop in the locations of the observed toe and artery floor intimal thickening [2]. The toe (T), heel (H), proximal (P), and distal (D) regions of the artery-graft junction are noted in Fig. 3, as are the laminar flow (LF) and stagnation point (SP) flow features. It is hypothesized that the toe and floor regions of vascular graft intimal thickening correspond to regions of flow oscillation and relatively low wall shear stress. Presumably, with adequate knowledge of the hemodynamic consequences of surgical procedures, improved designs or techniques can be developed to minimize flow disturbances, and thus maximize long-term graft patency.

Within the last two decades, as evidenced by the preceding two examples, detailed experimental in-

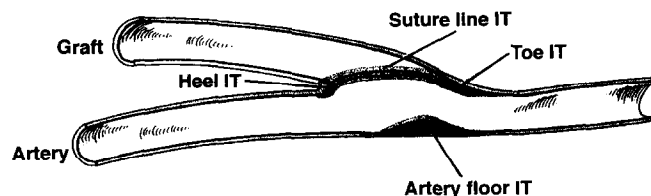


Fig. 2. Intimal thickening in an end-to-side vascular graft. (From Bassiouny et al. [2].)

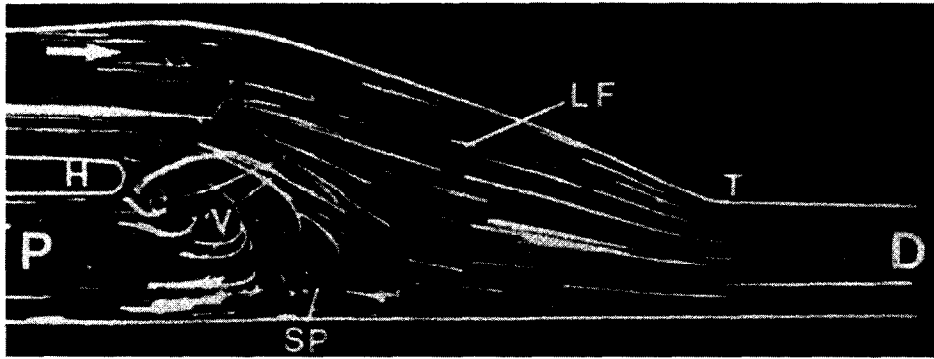


Fig. 3. Flow visualization studies in an end-to-side graft model. (From Bassiouny et al. [2].)

vesugations into the mechanics of blood flow in the vascular system have made significant contributions to the understanding of vascular disease and the long-term consequences of surgical repair. However, experimental model flow studies have several limitations. First, constructing the glass or silicone models, performing the experiments, and extracting quantitative flow data is often a time consuming and expensive process. Second, it is not feasible to use working fluids with the same complex rheological characteristics as blood, or models with the same mechanical and transport properties as blood vessels. Lastly, the quantitative information which can be extracted from experimental flow studies is usually limited to a small number of velocity profiles or pressure measurements. Complete field variations of scalar, vector, and tensor variables such as pressure, velocity, and viscous stress are impractical to obtain.

In recent years, computational techniques have been used increasingly by researchers seeking to understand vascular hemodynamics. Numerous two-dimensional analyses have been performed whereby arteries have been approximated as branched channels. These approximations, although constituting a necessary first step in the development of the computational framework, are largely inapplicable to arterial flow and will not be considered further. In contrast to the large number of two-dimensional solutions, few solutions to the three-dimensional, transient, flow equations governing blood flow have been performed. Of particular interest is the work of Perktold et al. [34] and the work of Ethier et al. [39] who are investigating pulsatile flow in three-dimensional models of bifurcations and bypass grafts.

Software and hardware technologies are emerging which will enable investigators to create idealized and patient-specific vascular models quickly, and to perform analyses combining accurate rheological models of blood, and constitutive relations and mass transport properties for blood vessels. Modern scientific visualization methods enable the immediate extraction of complete field variations of fluid and solid mechanical quantities. Precisely controlled 'computational experiments' with immediate data extraction and reduction, are an ideal means to test hypotheses of vascular function and augment information obtained from imaging, morphologic and molecular biologic methods.

In order to realize the tremendous benefits which can be derived from the application of computational methods in vascular medicine, several barriers have to be overcome. First, it is classically a time consuming and arduous process to create computer models of the complexity required for hemodynamic modeling in clinically relevant vascular systems. Second, generating a suitable finite element mesh from the continuous geometric model without suitable software tools is at best time consuming, and at worst, intractable. Third, the application of boundary conditions to large complex models can be difficult. Fourth, the discrete problems generated can involve the solution of hundreds of thousands, or millions of equations for thousands of time steps. Finally, the enormous amount of data generated must be reduced to the simplest possible form in order to extract meaningful clinical information.

The first major topic addressed in this paper is the development of a software system to integrate the model construction, mesh generation, problem initialization, three-dimensional pulsatile flow solution, scientific visualization, and quantitative data extraction aspects of computational vascular modeling. It is precisely this integration that enables the construction of models, solution of the governing equations, and extraction of relevant information for previously intractable problems such as flow in the abdominal aorta. The application of

this system to the analysis of blood flow in complex systems is the subject of the second major topic addressed. The computational method employed was validated with an analytic solution for pulsatile flow in a straight artery, and with experimental data for steady and pulsatile flow in an end-to-side vascular graft anastomosis and pulsatile flow in a carotid artery. Next, the computational method developed was used to quantitatively characterize the hemodynamic conditions under simulated rest and exercise pulsatile flow conditions in an idealized model of an abdominal aorta. Finally, the computational method developed was applied to compute the flow in patient-specific models created from pre-operative and post-operative medical imaging data of an aneurysmal abdominal aorta. This example illustrates the potential application of computational blood flow simulations to surgical planning.

The remainder of this paper is organized in the following manner: Section 2 presents some of the mathematical issues related to solving problems in vascular fluid mechanics. Section 3 presents the Stanford Virtual Vascular Laboratory, and the key developments and issues in system integration, modeling, mesh generation, finite element analysis, scientific visualization, and information extraction. Section 4 presents solved problems of increasing complexity, and comparisons between the computational solutions of blood flow with analytical solutions and experimental data. Section 5 presents conclusions and suggestions for future work.

2. Mathematical preliminaries

Computational modeling of blood flow requires solving, in the general case, three-dimensional, transient flow equations in deforming blood vessels. The appropriate framework for problems of this type is the arbitrary Lagrangian–Eulerian (ALE) description of continuous media in which the fluid and solid domains are allowed to move to follow the distensible vessels and deforming fluid domain [18,49].

The vessel diameter change during the cardiac cycle is observed to be approximately 5–10% in most of the major arteries, and, as a first approximation, the vessel walls are often treated as being rigid. In addition, in diseased vessels which are often the subject of interest, the arteries are even less compliant and wall motion is further reduced. Under the assumption of zero wall motion, the ALE description of incompressible flow in a deforming fluid domain reduces to the incompressible Navier–Stokes equations.

The strong form of the problem governing incompressible, Newtonian fluid flow in a fixed domain consists of the governing equations and suitable initial and boundary conditions. This problem is posed as follows for a domain $\Omega \subset \mathcal{R}^{n_{sd}}$ where n_{sd} is the number of spatial dimensions:

Given $\mathbf{f}: \Omega \times (0, T) \rightarrow \mathcal{R}^{n_{sd}}$, $\mathbf{g}: \Gamma_g \times (0, T) \rightarrow \mathcal{R}^{n_{sd}}$, $\mathbf{h}: \Gamma_h \times (0, T) \rightarrow \mathcal{R}^{n_{sd}}$ and $\mathbf{u}_0: \Omega \rightarrow \mathcal{R}^{n_{sd}}$. We seek $\mathbf{u}(\mathbf{x}, t)$, and $p(\mathbf{x}, t) \forall \mathbf{x} \in \Omega, \forall t \in [0, T]$ such that

$$\begin{aligned} \rho(\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u}) &= \nabla \cdot \boldsymbol{\sigma} + \mathbf{f} & (\mathbf{x}, t) \in \Omega \times (0, T) \\ \nabla \cdot \mathbf{u} &= 0 & (\mathbf{x}, t) \in \Omega \times (0, T) \\ \mathbf{u}(\mathbf{x}, 0) &= \mathbf{u}_0(\mathbf{x}) & \mathbf{x} \in \Omega \\ \mathbf{u} &= \mathbf{g} & (\mathbf{x}, t) \in \Gamma_g \times (0, T) \\ \mathbf{t} &= \mathbf{h} & (\mathbf{x}, t) \in \Gamma_h \times (0, T) \end{aligned} \quad (1)$$

where $\boldsymbol{\sigma} = -p\mathbf{I} + \mu \text{symm}(\nabla \mathbf{u})$ for a Newtonian fluid and $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$ is the traction vector. It is assumed that the initial velocity field, $\mathbf{u}_0$, is divergence-free.

The boundary conditions described above, involving the specification of velocity on part of the boundary and traction on the remaining part, results in a well-posed problem, but is not the only suitable combination of variables. Other well-posed boundary conditions involving other combinations of velocity, vorticity, traction, and pressure have been formulated. The well-posedness and applicability of different combinations of boundary conditions for the incompressible Navier–Stokes equations issue was recently addressed by Conca et al. [9] and Sani and Gresho [37]. Unresolved issues related to boundary conditions specification for vascular blood flow simulations will be discussed further in Section 5.

2.1. Womersley solution

Analytical solutions to the Navier–Stokes equations are possible in only a few select cases with very special geometry and boundary conditions. These solutions, when they exist, are essential for the validation of

numerical methods. However, most of these analytic solutions require the assumption of steady flow conditions and are thus of no use in addressing time accuracy issues for transient problems such as blood flow. Fortunately, a canonical solution for pulsatile flow does exist, the Womersley solution for fully developed pulsatile flow in a straight circular cylinder. To understand the assumptions required for the Womersley solution, we consider Fig. 4 where a long cylinder is subjected to a time varying flow rate at the inlet and a constant, but arbitrary, pressure over the outflow boundary. A sufficient distance from the inlet (which depends on geometric and flow parameters), the radial and circumferential components of velocity and pressure vanish, the axial velocity becomes a function of radius only, and the pressure varies linearly with axial position. Under these assumptions the Womersley solution can be derived.

In the case where the flow rate is known and the Womersley velocity profile is desired, a Fast Fourier Transform is used to extract the frequency content of the volume flow waveform given the fundamental frequency, ω . Thus, given $Q(t)$, the Fourier coefficients, B_n , are determined from

$$Q(t) \approx \sum_{n=0}^N B_n e^{in\omega t} \quad (2)$$

The Womersley velocity profile for the axial component of velocity, $w(r, t)$ is then given by

$$w(r, t) = \frac{2B_0}{\pi R^2} \left[1 - \left(\frac{r}{R} \right)^2 \right] + \sum_{n=1}^N \left\{ \frac{B_n}{\pi R^2} \left[\frac{1 - \frac{J_0\left(\alpha_n \frac{r}{R} i^{3/2}\right)}{J_0(\alpha_n i^{3/2})}}{1 - \frac{2J_1(\alpha_n i^{3/2})}{\alpha_n i^{3/2} J_0(\alpha_n i^{3/2})}} \right] \right\} e^{in\omega t} \quad (3)$$

Here, R is the radius of the cylinder, J_0 and J_1 are Bessel functions of the first kind of order 0 and 1, respectively, and $\alpha_n = R\sqrt{(n\omega)/\nu}$ where ν is the kinematic viscosity. The nondimensional parameter $\alpha = R\sqrt{\omega/\nu}$ is known as the *Womersley number*.

Fig. 5 displays the results of the procedure for a sinusoidal input flow waveform. Note that the radial velocity variation is not, in general, parabolic, and in fact reverses in direction close to the wall even though the total volume flow always remains positive. This boundary layer of reversed flow close to the wall is an important characteristic of pulsatile flow, and illustrates that even for a straight section of artery with positive volume flow, there is always some time in the cardiac cycle where the viscous traction forces are opposite the dominant flow direction.

The velocity profile function expressed in Eq. (3) was implemented as a user defined boundary condition function for the SpectrumTM Solver. Thus, given a node on a boundary where a Womersley velocity function is

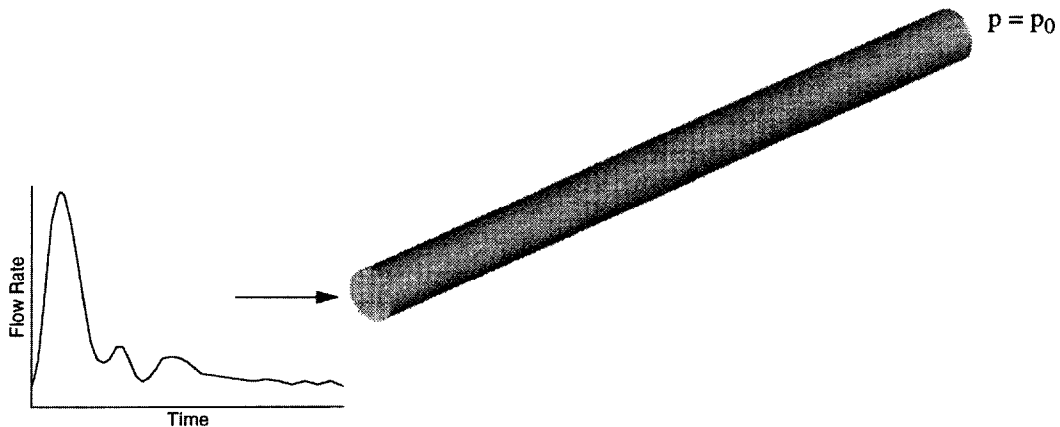


Fig. 4. Problem description for pulsatile flow in a long, straight cylinder.

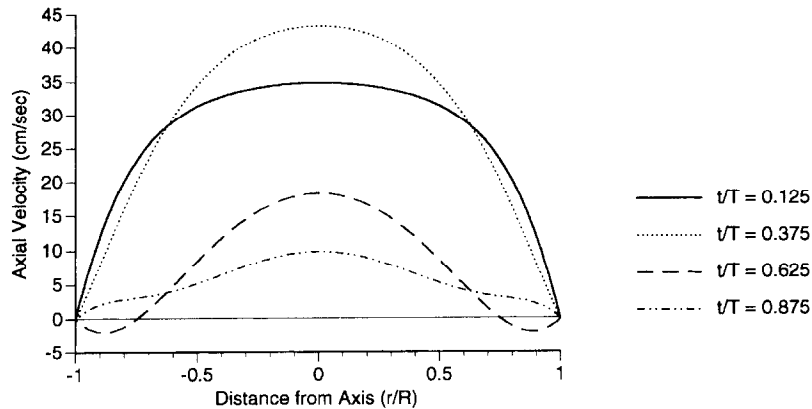


Fig. 5. Womersley solution for pulsatile flow in a long, straight cylinder.

prescribed, as well as the Fourier coefficients of the volume flow rate function, period, and fluid properties, the velocity at that node is computed and prescribed.

3. Virtual Vascular Laboratory

Within the context of the computer aided engineering design and analysis process, approximately 80–90% of an analyst's time is generally spent constructing geometric models, discretizing these models, and preparing input data for analysis programs. Researchers using computational tools to model biomechanical processes have encountered this same impediment to progress. With so much time being spent on tasks which have little or nothing to do with understanding the underlying biophysical phenomena, progress in computational modeling of blood flow has come relatively slowly. In addition, due to inadequate geometrical modeling and automatic mesh generation tools, the anatomical variation so common in the human vasculature is rarely incorporated into studies of blood flow. Researchers run the risk of drawing incorrect conclusions about vascular blood flow by not exploring the 'design space' of human anatomy.

This section presents an integrated software system, the Virtual Vascular Laboratory, for quickly creating complex geometric models, modifying these models to reflect anatomical variations, automatically generating finite element meshes, writing out problem data including boundary conditions and customized solution strategies, and then running a finite element solver and scientific visualizer. This integrated software approach has reduced the time it takes to create vascular blood flow computational problems to hours instead of weeks, and requires a minimum amount of user interaction.

2.1. System overview

Within the engineering community, model creation and discretization barriers are being surmounted through the utilization of software design tools often termed Simulation Based Design (SBD), or Computational Prototyping, systems. The idea is to use knowledge-based engineering principles to extract information about the physical system including geometry, material properties, and loads and constraints to create a parametric model incorporating that knowledge. An object-oriented framework ensures maximum extensibility and reusability of modeling information. The organization of an SBD system is depicted in Fig. 4.

Knowledge-based engineering software provides the 'glue' to integrate all of the other software technologies. It incorporates an object-oriented environment to create a unified model combining data input from computer aided design (CAD) or medical imaging sources, geometry, mesh generation techniques, solution strategies, and visualization techniques within a common graphical user interface. Geometric solid models can be constructed directly from primitives such as lines, surfaces, or solids, or by extracting anatomical surfaces or features from medical imaging data sets obtained from computed tomography (CT), magnetic resonance imaging (MRI), or other sources. Once constructed, these models must be easily modified to simulate a wide array of possible

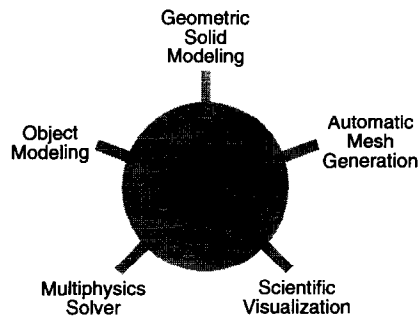


Fig. 6. Enabling technologies for simulation-based design.

anatomical variations or surgical procedures. Parametric design principles and geometric computing operators facilitate this process. Automatic mesh generation strategies, operating directly on the geometric solid model, minimize the time spent by the biomedical analyst in discretizing the continuous problem domain. Solution methods capable of simultaneously modeling the multiple biophysical processes such as fluid flow, solid deformation, and mass transport occurring in the human body are essential. Further, due to the large problem sizes inherent in three-dimensional, transient blood flow modeling, these methods should exploit parallel architectures finding increasing use in scientific computing. Scientific visualization methods must include techniques for viewing various scalar, vector, and tensor quantities of these multiple bio-physical phenomena. These techniques must maintain the generality necessary to model different phenomena yet must be easily specialized to maximize insight into an individual problem. Finally, a graphical user interface must allow individuals to interact with the modeling environment using terminology and concepts specific to their disciplines.

Complex biophysical problems often require the solution of extremely large systems of nonlinear evolution equations. Within the context of biomechanical applications, workstations are currently most widely used for

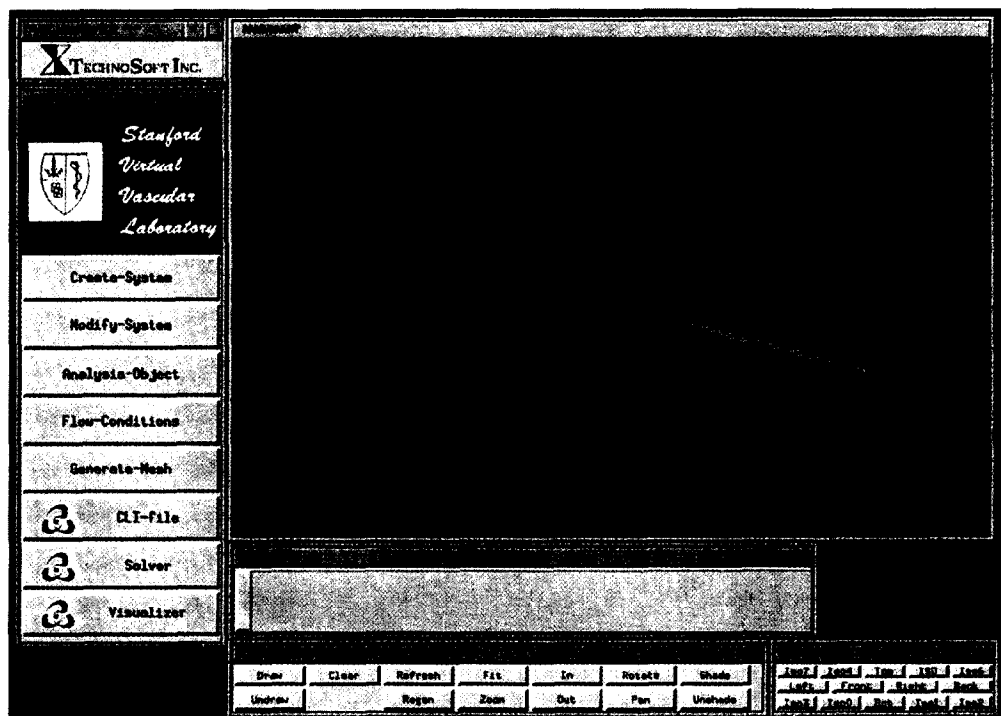


Fig. 7. User interface to the Stanford Virtual Vascular Laboratory.

creating geometric models, mesh generation, solving the system of equations and visualizing the results. Due to large problems sizes, parallel architectures will be the systems of choice for most of these tasks in the near future, thus imposing an additional requirement on software components.

The Virtual Vascular Laboratory is a software environment, modeled after simulation based design systems in use in industrial applications. The graphical user interface created for this system is shown in Fig. 7. This system was designed to incorporate functionality to create an object model from a list of models including carotid arteries, bifurcations, abdominal aorta, thoracic aorta, end-to-side vascular grafts, and others. All of the models are parametrically defined and hence all of the parameters can be changed to create a new artery model. Analysis objects can be selected to be a subset of the anatomic model. For example, in many cases, only one half of a symmetric model needs to be discretized and used for blood flow computations. In addition to the flow models considered in the present work, methods to simultaneously create a consistent geometric model of the blood vessel wall were implemented. Boundary conditions are defined on the geometric model and are thus automatically updated when this geometry changes. Full functionality exists to initialize the pulsatile flow boundary conditions given a volume flow waveform. The appropriate region to mesh is automatically selected and default mesh refinement parameters specified based on local vessel sizes. Local changes in the mesh refinement can be specified to further resolve features in a particular region of interest such as in the neighborhood of branches. The complete set of input files for the finite element solver, SpectrumTM, are generated automatically. Finally, the finite element solver itself can be started and, upon completion of the analysis, the scientific visualizer employed.

Fig. 8 depicts the construction of a solid model from geometric primitives, consisting of circles oriented along

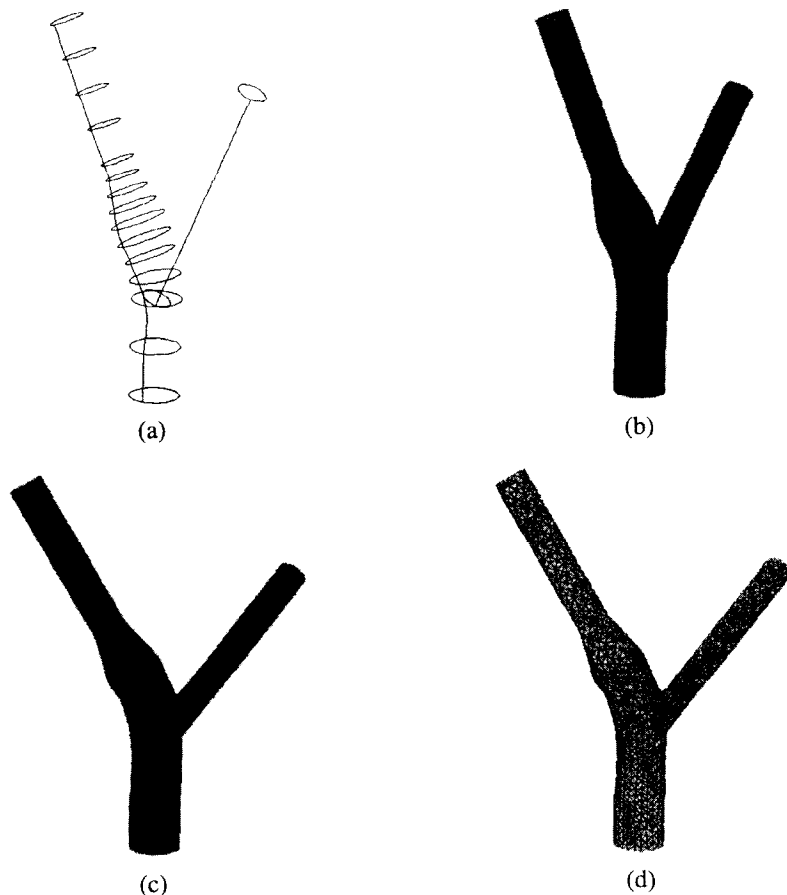


Fig. 8. Geometry construction, modification and mesh generation example showing: (a) circular cross-sections defining original geometric model; (b) original geometric solid model with branch angle of 45°; (c) modified geometric solid model with branch angle of 60°; (d) finite element mesh of one half of the modified model.

a path in space, subsequent modification of the model to reflect a larger branch angle, and generation of a finite element mesh on one half of the model. Other vascular segments including an arbitrary number of branches are constructed and discretized with the same approach.

The key software components used in the Virtual Vascular Laboratory are shown in Fig. 9. Although other software programs could replace the component parts, the ones chosen were, in the authors' opinions, the most suitable for the present work.

The *Adaptive Modeling Language*TM (AML), is the knowledge-based engineering and object modeling software used to create the integrated system [47]. SHAPESTM is incorporated within AML and use as the geometry kernel [48]. The automatic finite element mesh generator, Finite OctreeTM, is used to generate the finite element discretizations from the solid model [38]. Input files are automatically created in the Virtual Vascular Laboratory for the finite element analysis program, SpectrumTM, which supports the multiphysical computations typifying vascular mechanics [49]. User defined functions provide the necessary capability to include pulsatile flow boundary conditions, rheological models for blood, and constitutive equations for blood vessels. Parallel computers are used for the solution of the nonlinear evolution equations governing blood flow. The SpectrumTM Visualizer is utilized to view the results of the computations and extract quantities of interest including pressure and velocity fields, wall shear stress distributions, and particle residence times [50].

3.2. Knowledge-based engineering

Knowledge-based engineering (KBE) principles form the central element of the Virtual Vascular Laboratory. These principles, and software, allow for the integration of the different component technologies required to solve complex vascular blood flow problems. Different classes of knowledge are first described, followed by a discussion of how KBE methods were used to orchestrate the sequence of events necessary for creating and modifying computational models of vascular blood flow.

3.2.1. Introduction

Knowledge-based engineering systems can be defined, in the present context, as computer software developed to encapsulate heuristic knowledge of an engineering problem domain assist humans in performing a specified intellectual task [12]. The knowledge to be encapsulated includes the form, function, and behavior of the problem domain. As noted in [13] this knowledge can be classified as (1) first principles, e.g. the equation of conservation of mass, (2) phenomenological models, e.g. viscosity constitutive models, (3) analytical models, e.g. the telegrapher's equation for the propagation of the pressure pulse wave, (4) numerical representations, e.g. a discretized model and strategy for the solution, or (5) heuristics, e.g. rules governing branch ratios of blood vessels or design constraints for vascular grafts. It is clear from the preceding examples that a wide array of information, incorporating all of the above classifications of knowledge, is required to solve engineering problems in general and vascular blood flow problems in particular. The role of the knowledge-based system is to provide a framework for organizing and utilizing this problem specific information.

3.2.2. Implementation

Knowledge-based engineering methods implemented in AMLTM are used in the Virtual Vascular Laboratory to organize the multiple levels of information required for the solution of problems of vascular blood flow.

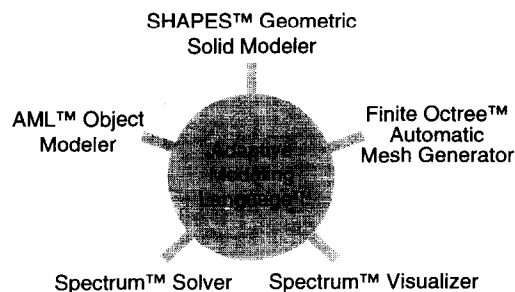


Fig. 9. Software used in the Stanford Virtual Vascular Laboratory.

These include physical information about the vascular anatomy such as the relationships between the dimensions of the blood vessels. A specific example is the fact that all of the dimensions of an idealized model of a carotid artery are based on the nominal diameter of the common carotid artery. This allows for the immediate scaling of the anatomy while maintaining a topologically similar model. Any dependent quantities such as velocity input conditions are updated automatically by the system functions, which manage the relationships and constraints amongst the vascular objects and propagate the constraints throughout the model. Using demand-driven and dependency backtracking techniques, only objects or properties which depend on a modified property are updated, significantly reducing model regeneration requirements.

In summary, the computational solution of vascular blood flow problems requires the integration of knowledge of different forms including phenomenological, analytical, numerical and heuristic knowledge. Further, the modification of parameters defining the model often triggers a complex sequence of events resulting, ultimately, in a new problem to be solved. The role of knowledge-based engineering software, in the present context, is to manage the numerous model representations, relationships, and constraints defining a vascular blood flow problem in order to simplify the process of interacting with, and changing, the problem.

3.3. Object modeling

The representation of the data of the model is independent of the knowledge-based environment for managing it. However, certain data representations are easier to manage and incorporate into the overall system. An object-oriented approach to software development is akin to the use of interchangeable parts in manufacturing, clearly defining the relationships amongst the components and facilitating the incorporation of additions and changes as the system evolves. This approach is presented and the relevance to problems of vascular blood flow described. The development of the object model used in the present work is then detailed.

3.3.1. Introduction

The human vascular system is a complex system comprised of billions of blood vessels of different form and function. Fortunately, as noted by Simon [40], ‘many complex systems have a nearly decomposable, hierarchic structure’ enabling us to more easily understand the system. This decomposable, hierarchic structure has provided the impetus for the development of object-oriented methods utilized in recent years in design, analysis, and computer programming. The object-oriented paradigm, and its applicability to vascular modeling is first described, followed by an examination of the issues related to the implementation of object models for vascular problems.

The central element of an object-oriented approach, the object model, has four major elements as identified by Booch [5]:

- (1) Abstraction
- (2) Encapsulation
- (3) Modularity
- (4) Hierarchy

Abstraction, in the present context, refers to an idealized form of a system or process. For example, when investigating hemodynamics, a segment of a blood vessel can be abstracted as being comprised of a thin-walled elastic tube filled with a Newtonian fluid. This abstraction might be appropriate for investigations of the effect of the deformability of blood vessel on hemodynamic variables. Alternatively, a blood vessel can be abstracted as consisting of lumped resistive and capacitive elements for use in an electrical analog model of the vasculature. The essential characteristics, and consequently the abstraction, of the object or system clearly depend upon the particular question to be investigated and the techniques used to address that question.

Encapsulation refers to the separation of the internal details of one part of an object model from the other parts, thereby easing the process of changing a single component. For example, when investigating the blood flow in the carotid artery bifurcation, the precise nature of the blood flow patterns in the abdominal aorta is unimportant; these flow patterns could be encapsulated into the net effect of the abdominal aorta on systemic conditions. Thus the influence of the abdominal aorta on carotid artery flow is manifested only at the boundaries of the carotid artery model.

Modularity refers to the property of a system which can be partitioned into its component parts. In the vascular system, by naming the individual arteries we are noting that these vessels, although inter-connected, are

conceptually modular, and hence can be considered individually. As a further example of modularity, a vascular graft can be thought of as an independent component, with its own geometric and material properties, which can be incorporated into many different arterial subsystems.

Hierarchy is defined in the present context by Booch [5] as ‘a ranking or ordering of abstractions.’ Courtois [10] describes the hierarchy of complex systems in a way that is particularly relevant for the vascular system:

“Frequently, complexity takes the form of a hierarchy, whereby a complex system is composed of interrelated subsystems that have in turn their own subsystems, and so on, until some lowest level of elementary components is reached.”

The human vascular system, comprised of blood vessels with varying compositions, shapes and sizes, can be organized hierarchically in terms of arteries, arterioles, capillaries, veins, and venules. Additionally, a single vascular segment can be organized hierarchically in terms of the interior fluid and the vessel wall. The vessel wall can be organized further into the endothelial layer, the intima, the media and the adventitia. It is clear that this organization can proceed indefinitely from coarse to fine granularity depending upon the requirements of a specific investigation. Further, a hierarchical approach is a natural method to organize a model of the vascular system.

It is seen from the preceding paragraphs that the human vascular system is conceptually representable within an object-oriented framework. However, the implementation of this approach in a software environment requires the development of vascular objects incorporating relevant anatomic and physiologic data.

3.3.2. Implementation

An object model of an arbitrary arterial blood flow problem for the human vascular system was developed using the Adaptive Modeling LanguageTM [47]. The arterial blood flow problem object is general and can be specialized for individual studies of the vascular system such as flow in the carotid artery bifurcation, abdominal aorta and major branches, or even larger arterial systems. Fig. 10 displays the object tree structure for a vascular

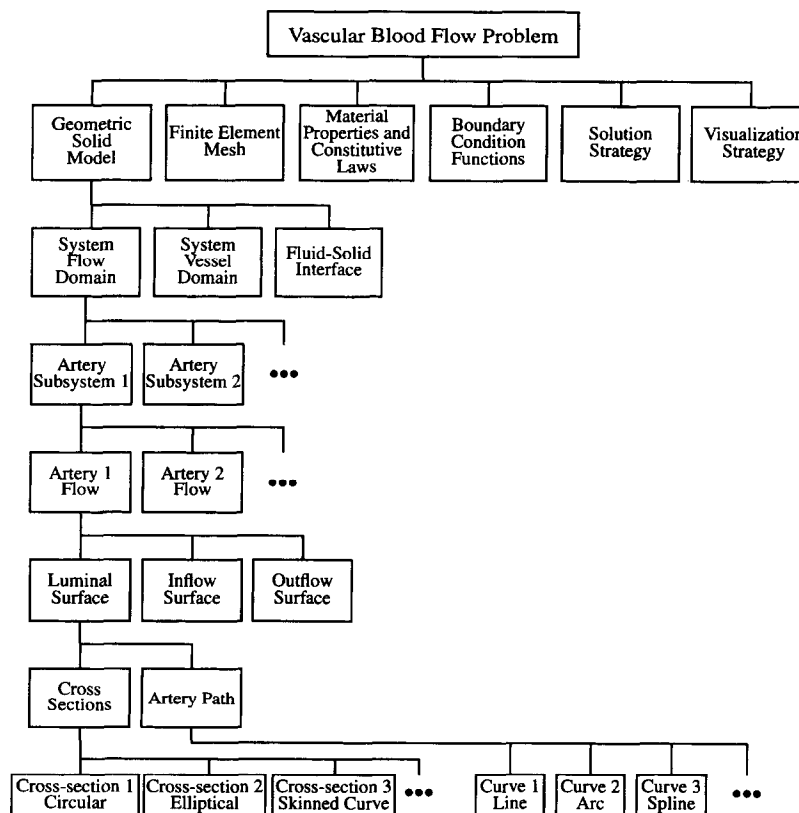


Fig. 10. Organization of the object model for vascular blood flow problems.

blood flow problem. At the highest level, separate objects are created for the geometric model, finite element mesh, material properties and constitutive laws, boundary condition functions, solution strategy, and visualization strategy. The anatomy of an arterial subsystem is combined with the relevant physiologic information, specified as boundary conditions and material properties, and knowledge about mesh generation and analysis procedures to form a blood flow problem. This knowledge is linked together by constraints, managed by AML™, so that when, for example, the anatomic configuration is changed, any dependent information related to mesh generation, boundary conditions, or analysis methodology is updated by dependency backtracking.

The subobjects contained in the geometric solid model, shown in Fig. 10, are illustrated in detail as this is directly related to the anatomic structure of an arterial system. The geometric solid model is composed of the system flow domain, the system vessel domain, and the surface representing the interface between the domains, the luminal surface. The flow domain is the only domain required for the problems presented in this paper, and therefore, will be detailed further. Note that the capabilities for blood flow/vessel deformation problems are functional and have been used to solve coupled problems, but will not be reported upon herein.

The total system flow can be further decomposed into that of subsystems. For example, the thoracic aorta and abdominal aorta would be the two subsystems required in an investigation of flow in the entire aorta. Each subsystem is further decomposed into its component artery objects. Thus, the flow region for a study of the carotid bifurcation is the union of the flow domains of the common carotid artery, internal carotid artery, and external carotid artery. Each of these flow domains is defined as the region bounded by the luminal, inflow, and outflow surfaces. The luminal surface is then decomposed in terms of a collection of cross-sections and a path along which these cross-sections are oriented. Then, finally, at the lowest level of abstraction, each of the cross-sections can be circular, elliptical, or defined as a curve through a set of points. Additionally, the artery path can be defined by a sequence of lines, arcs, splines, or other curves.

It is important to note that the object-oriented architecture underlying the vascular blood flow problem does not change for different problems. In the terminology of object modeling, a class has been created which can be instantiated to create different vascular blood flow problems by specializing the overall structure to a specific instance.

As an example of the utility of this approach and the economies possible, Fig. 11 illustrates how a canonical bifurcation object, once defined, can be used to create a system of bifurcations. This is performed by simply creating copies of the canonical object and establishing a constraint on the dimensions and position of a child bifurcation to coincide with the dimensions and positions of its parent. Successive generations of vessels are created in the same fashion. Subsequently, the geometric model of the composite structure is created as the union of the component parts, the inflow, outflow, and vessel wall boundaries are identified, and a new problem created.

The organization of the anatomic data for the vascular subsystem further illustrates the object-oriented

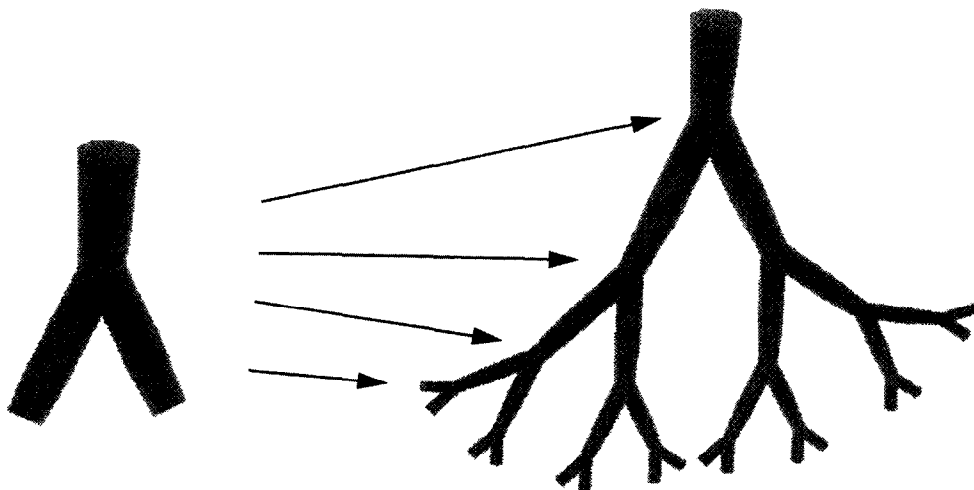


Fig. 11. Object model example displaying canonical bifurcation and system of bifurcations.

approach to vascular modeling. Two methods were implemented to define the anatomic model. First, as described above, by defining arteries individually and assembling them to form a subsystem, and second, by extracting the luminal surface from medical imaging data and constructing the subsystem directly from this surface. The object structure detailed in Fig. 10 remains exactly the same down to the level where the subsystem is defined. The first method, the individual definition of the arteries, is suitable for the construction of idealized models, as shown in Fig. 11, whereas the second is more appropriate for creating patient-specific models. Fig. 12 illustrates the construction of a bifurcation model from a surface of constant velocity magnitude extracted from phase contrast magnetic resonance imaging data. A hybrid between the idealized and patient-specific approaches is appropriate in the case where a graft model is added to a patient-specific anatomic model.

3.4. Geometric modeling

3.4.1. Introduction

One of the most important components of a virtual vascular laboratory, or simulation based design system in general, is the geometric representation. A geometric modeler is essential for creating an analytical description of an object from geometric primitives including points, arcs, surfaces, and solids as well as derived objects such as surfaces extruded from curves, or solids bounded by surfaces. The geometric modeler performs the function of creating and maintaining a coherent system geometry from multiple components.

3.4.2. CAD approach to solid model construction

In the computational biomechanics field, traditional CAD-based approaches have been the primary means of constructing computational solid models. In regards to constructing models of vascular anatomy, primitive forms such as circles and ellipses can be oriented in space and skinned to form a surface. These surfaces can be capped on the ends and the resulting surfaces used to define an enclosed, solid region. In this way, a model of the interior of a single artery can be formed. Multiple arteries, formed in this fashion, can be created in local spaces, oriented and then connected together to construct an arterial subsystem. However, these connections must be formed in such a way as to ensure that when the geometric model is discretized by a finite element mesh generator, a consistent mesh is obtained. Namely, free edges and surfaces must be eliminated from the interior portions of the mesh.

The creation of a coherent model requires geometric boolean operations such as union, difference, trim, and intersection and hence requires a geometric solid modeler. Further, non-manifold topology must be supported to



Fig. 12. Bifurcation object defined by extracting a surface of constant velocity magnitude from magnetic resonance imaging data.

allow for the possibility of solids, surfaces and curves in the same model. A simple example in arterial flow where this would be required is presented by the case of modeling the inclusion of a stent-graft (graft approximated as a thin shell, stent approximated by beams) inside of a flow stream described by a solid region. The geometric modeler utilized in the present work, SHAPESTM, supports the boolean operations and non-manifold topology necessary for modeling complex vascular biomechanics problems.

3.4.3. Webbed surface constructions of solid models

Solid models can also be constructed with bounding surfaces comprised of a set of triangular or quadrilateral facets. As noted previously, triangular facets are created by surface extraction techniques widely used in the medical imaging field. Decimation algorithms can be used to reduce the number of facets and significantly reduce the time required for rendering and geometric operations. Smoothing algorithms have been recently described by Lorensen [28] to reduce artifacts introduced in the imaging process.

3.5. Automatic mesh generation

3.5.1. Different strategies for mesh generation

In recognition of the fact that one of the most difficult aspects of using computational techniques to solve partial differential equations is the generation of the discretization, significant resources have been devoted to develop automatic mesh generators. For complex problems, such as blood flow in multi-branched arterial systems like the abdominal aorta, this is especially difficult and precludes the use of simple ‘structured’ mesh generators. The generation of fully unstructured finite element meshes for arbitrarily complex geometry has become the ‘holy grail’ of the computational mechanics community, spawning large mesh generation consortiums world-wide.

Of the mesh generation strategies currently available, only the octree and advancing front techniques have demonstrated the capability to generate meshes for complex vascular flow models. The Finite OctreeTM mesh generator developed by Shephard et al. [38] at Rensselaer Polytechnic Institute was used to generate the complex, unstructured meshes appearing in this paper and is described subsequently.

3.5.2. Finite OctreeTM method

The basic idea behind the octree method lies in placing the geometric object to be meshed in a cube which completely contains the object. This cube is recursively subdivided into octants of finer and finer scale depending on the mesh size parameters of the object, specified by the user. Octants which are inside the object are divided into tetrahedral elements, those outside the object are excluded. Octants which contain a portion of the object are trimmed to match the object surface, to a given tolerance level, and the subdivided into tetrahedral elements. Following discretization, mesh smoothing operations are performed to improve element quality.

A key feature of the present approach to mesh generation is the ability to assign different elements sizes to different objects. For example, a maximum element size of 1 mm might be suitable for discretizing the aorta, which is approximately 25 mm in diameter at the level of the diaphragm, but would not be appropriate for the inferior mesenteric artery, which is on the order of 3 mm in diameter. Consequently, in the set of information that is passed to the mesh generator, the maximum element size of an artery region is included. Typical discretizations include a minimum of 10 elements across an individual artery cross-section. It should be noted that, due to the underlying knowledge-based architecture of the Virtual Vascular Laboratory, mesh refinement parameters are automatically assigned a default value, based on a nominal artery diameter and a global refinement parameter. The global refinement parameter can be adjusted to yield coarse, medium, or refined discretizations.

3.6. Automatic generation of analysis problem data

One of the distinguishing features of knowledge-based engineering systems is the ability to keep track of problem-specific data in addition to a geometric model and, further, to link this problem data directly to the geometric model such that when this model is changed, the data is updated automatically. Thus, in the hierarchical model created for vascular modeling, a distinct analysis model is created in addition to the geometric model, whereby the analysis model incorporates simplifications to the geometric model, whereby the

analysis model incorporates simplifications to the geometric model, e.g. problem symmetry, as well as information specific to the analysis to be performed.

Boundary condition data is typically defined on the geometry, but specified on the discretization. For example, in blood flow computations, the nodes along the luminal surface of a vascular system are specified to have zero velocity in accordance with a no-slip boundary condition. Thus, the luminal surfaces of all of the blood vessels comprising this arterial system first need to be identified. This identification is independent of any possible discretization. After a finite element mesh is generated and analysis problem is identified, the finite element nodes and elements along the luminal surface are identified by querying the mesh to determine the correspondence between the finite element mesh and the model topology. The subset of nodes and element faces along the luminal surface are identified and grouped into a boundary condition object.

3.7. Finite element solution methodology

The finite element method is employed to create a finite-dimensional approximation to the equations governing blood flow in arteries. The SpectrumTM Solver was used together with user-defined functions for boundary conditions to solve this finite dimensional system of equations. Due to the fact that large systems of equations are created for relevant blood flow computations, an iterative solution strategy was utilized to reduce memory requirements and the larger computations were performed on parallel computers. The finite element method and solution strategy employed is discussed in more detail below.

3.7.1. Stabilized methods for incompressible Navier–Stokes

Stabilized finite element methods were introduced by Hughes and colleagues [6,19,21] to correct deficiencies in the standard Galerkin finite element method in advection dominated flows. These methods have been widely applied to solve compressible and incompressible fluid mechanics problems on scalar, vector and parallel computers.

The essential features of stabilized methods in the context of incompressible flows are the simultaneous stabilization of the advection operator and the circumvention of the Babuska–Brezzi inf–sup condition restricting the use of many convenient interpolations, including the linear velocity, linear pressure interpolations used in the present work. The basic idea is to augment the Galerkin finite element formulation with a least squares form of the residual, including appropriate stabilization parameters. These stabilization parameters are designed so that the method achieves exact solutions in the case of one-dimensional model problems involving, for example, the steady advection–diffusion equation.

Stabilization methods have been used with both semi-discrete and space–time formulations. Space–time formulations have been widely employed in recent years in the computational fluid mechanics field, but have not yet reached the same level of sophistication for solid mechanics problems where semi-discrete methods are the norm. The present formulation is semi-discrete in time due to the necessity of coupling with existing solid mechanics formulations for the more general class of problems involving flow through deforming blood vessels.

The stabilized method employed in the present work is described as follows. We start with the mathematical problem of incompressible flow stated in Section 2 and define the finite element function spaces

$$\begin{aligned} V_h &= \{v(\cdot, t) \in [H_0^1(\Omega)]^{n_{sd}} | v|_K \in [R_1(K)]^{n_{sd}} \forall K \in T_h, v(\cdot, t) = \mathbf{0} \text{ on } \Gamma_g \times (0, T)\} \\ S_h &= \{u(\cdot, t) \in [H_0^1(\Omega)]^{n_{sd}} | u|_K \in [R_1(K)]^{n_{sd}} \forall K \in T_h, u(\cdot, t) = g \text{ on } \Gamma_g \times (0, T)\} \\ P_h &= \{p(\cdot, t) \in C_0(\Omega) \cap L_0^2(\Omega) | p|_K \in R_1(K) \forall K \in T_h\} \end{aligned} \quad (4)$$

where $R_1(K)$ is the first order polynomial space on K .

The semi-discrete Galerkin method is: Find $u \in S_h$ and $p \in P_h$ such that

$$\begin{aligned} \forall w \in V_h, q \in P_h \\ B(w, q; u, p) &= 0 \\ B(w, q; u, p) &= \left(w, p \frac{\partial u}{\partial t} + \rho u \cdot \nabla u \right)_\Omega - (\nabla \cdot w, p)_\Omega + (\nabla w, \mu \nabla u)_\Omega - (\nabla q, p u)_\Omega \\ &\quad - (w, h)_\Gamma - (w, f)_\Omega + (q, \rho u_n)_\Gamma \end{aligned} \quad (5)$$

With Eq. (5) as a point of departure we define our stabilized method as

Find $\mathbf{u} \in S_h$ and $p \in P_h$ such that $\forall \mathbf{w} \in V_h, q \in P_h$

$$B_S(\mathbf{w}, q; \mathbf{u}, p) = 0$$

$$\begin{aligned} B_S(\mathbf{w}, q; \mathbf{u}, p) = & B(\mathbf{w}, q; \mathbf{u}, p) + (\mathbf{w}, \rho(\bar{\mathbf{u}} - \mathbf{u}) \cdot \nabla \mathbf{u})_{\Omega} \\ & + \left(\rho \mathbf{u} \cdot \nabla \mathbf{w} + \rho \nabla q, \tau_1 \left(\rho \frac{\partial \mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p - \mathbf{f} \right) \right)_{\Omega} \\ & + (\rho \nabla \cdot \mathbf{w}, \tau_2 \nabla \cdot \mathbf{u})_{\Omega} + (\rho(\bar{\mathbf{u}} - \mathbf{u}) \cdot \nabla \mathbf{w}, \tau_3 (\rho(\bar{\mathbf{u}} - \mathbf{u}) \cdot \nabla \mathbf{u}))_{\Omega} \end{aligned} \quad (6)$$

with

$$\tau_1 = \frac{C}{\sqrt{c_1 \rho^2 u_i g_{ij} u_j + c_2 \nu^2 g_{ij} g_{ij} + c_3 \frac{\rho^2}{\Delta t^2}}} \quad (7)$$

$$\tau_2 = \frac{\sqrt{u_i g_{ij} u_j}}{\rho \operatorname{tr}(g_{ij})} \quad (8)$$

$$\tau_3 = \frac{C}{\sqrt{c_1 \rho^2 (u_i - \bar{u}_i) g_{ij} (u_j - \bar{u}_j)}} \quad (9)$$

and

$$\bar{\mathbf{u}} = \mathbf{u} - \tau_1 \rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \frac{\nabla p}{\rho} - \mathbf{f} \right) \quad (10)$$

where $\tilde{\Omega}$ is the restriction of Ω to element interiors, g_{ij} is the covariant metric tensor, C is a constant depending on the element topology, and c_1 , c_2 and c_3 are constants defined from the one-dimensional scalar model problem. A second-order accurate time-stepping algorithm is introduced into the above semi-discrete nonlinear description resulting in a nonlinear algebraic problem. This nonlinear problem is linearized and the resulting linear system of equations is iteratively solved to convergence within a time step.

3.7.2. Staggered solution strategy

The system of linear equations for the increments in velocity and pressure can be written as

$$\begin{bmatrix} \mathbf{K} & \mathbf{G} + \mathbf{D}_u \\ \mathbf{G}^T + \mathbf{D}_p & -\mathbf{C} \end{bmatrix} \begin{Bmatrix} \Delta \mathbf{u} \\ \Delta p \end{Bmatrix} = \begin{Bmatrix} -\mathbf{R}_u \\ \mathbf{R}_p \end{Bmatrix} \quad (11)$$

where $\mathbf{K}$ is a matrix including contributions from inertial, advective and diffusive terms in Eq. (6), $\mathbf{G}$ is the gradient operator, $\mathbf{G}^T$ is the divergence operator, $\mathbf{C}$, $\mathbf{D}_u$ and $\mathbf{D}_p$ arise from the stabilization terms, and $\mathbf{R}_u$ and $\mathbf{R}_p$ are the residuals arising from the momentum and continuity equations, respectively. Eq. (11) is a nonsymmetric system not amenable to efficient iterative solution. A symmetric form is obtained by neglecting the nonsymmetric terms and is written as

$$\begin{bmatrix} \tilde{\mathbf{K}} & \mathbf{G} \\ \mathbf{G}^T & -\mathbf{C} \end{bmatrix} \begin{Bmatrix} \Delta \mathbf{u} \\ \Delta p \end{Bmatrix} = \begin{Bmatrix} -\mathbf{R}_u \\ \mathbf{R}_p \end{Bmatrix} \quad (12)$$

where $\tilde{\mathbf{K}} = (1 + \alpha) \operatorname{diag}(\mathbf{K})$ and α is a positive regularization parameter.

We can use Eq. (12) to express $\Delta \mathbf{u}$ in terms of Δp as

$$\Delta \mathbf{u} = -\tilde{\mathbf{K}}^{-1} (\mathbf{R}_u + \mathbf{G} \Delta p) \quad (13)$$

Symbolically, we have obtained an equation for Δp :

$$(\mathbf{C} + \mathbf{G}^T \tilde{\mathbf{K}}^{-1} \mathbf{G}) \Delta p = -\mathbf{R}_p - \mathbf{G}^T \tilde{\mathbf{K}}^{-1} \mathbf{R}_u \quad (14)$$

This is a symmetric system of linear equations in the form of a discrete Poisson problem. After solution, we

update $\mathbf{u}$ and p and then compute $\Delta \mathbf{u}$ by solving a regularized advection–diffusion equation using the Generalized Minimal Residual (GMRES) algorithm introduced by Saad and Schultz [36], namely

$$(\mathbf{K} + \beta \mathbf{K}_r) \Delta \mathbf{u} = -\mathbf{R}_u - (\mathbf{G} + \mathbf{D}_u) \Delta p \quad (15)$$

where β is a regularization parameter and $\mathbf{K}_r$ is a diagonal regularization matrix based on the stabilization parameters. Finally, we update $\mathbf{u}$ by $\mathbf{u} + \Delta \mathbf{u}$ and p by $p + \Delta p$ to complete a stagger. This process is continued (usually for 2–4 staggers) until an acceptable level of convergence is obtained. The solution is then advanced to the next time step. This staggered solution algorithm results in a robust method of solving incompressible flow problems. Typically, the velocity stagger converges in less than 5 GMRES iterations and the pressure stagger converges in approximately 20 Conjugate Gradients iterations.

3.7.3. Parallel computation issues

Three-dimensional transient solutions of blood flow in arteries impose severe requirements on computer software and hardware architecture in memory storage and computational speed. Consider the finite element discretizations (Fig. 13) of idealized and patient-specific abdominal aorta models. Each of these finite element meshes has in excess of 1/4 million nonlinear equations to be solved for hundreds of time steps.

Explicit methods are widely used for solving large problems in fluid and solid mechanics for small time intervals. However, for many problems of fluid mechanics an implicit method is preferred due to the much larger time step sizes which can be used. Iterative solution methods such as GMRES or Conjugate Gradients are generally preferred for large problems due to the excessive memory requirements of direct solvers where a large matrix must be stored and factorized. However, even with iterative equation solvers, memory requirements are often excessive. A matrix-free GMRES solver developed by Johan et al. [22] and incorporated in the SpectrumTM Solver, was utilized for the parallel solutions presented, substantially reducing memory requirements to the point where solution time becomes the limiting factor in problem size.

The maximum problem size considered herein involves approximately 250 000 nonlinear equations solved for approximately 200–500 time steps per cardiac cycle for a total of 3 cycles. The use of low memory, parallelized computer software algorithms on parallel architectures has enabled the solution of problems one to two orders of magnitude larger than previously reported computations of vascular blood flow. These problems are discussed in detail in the next section.

3.8. Scientific visualization and information extraction

The extraction of relevant physiologic information from the computational analyses poses significant challenges in scientific visualization. Experimental fluid mechanics techniques such as laser doppler anemometry might involve at best hundreds of data values of a single field quantity to record and examine, yet in the computational studies presented in the present work, scalar, vector, and tensor fields are computed at hundreds

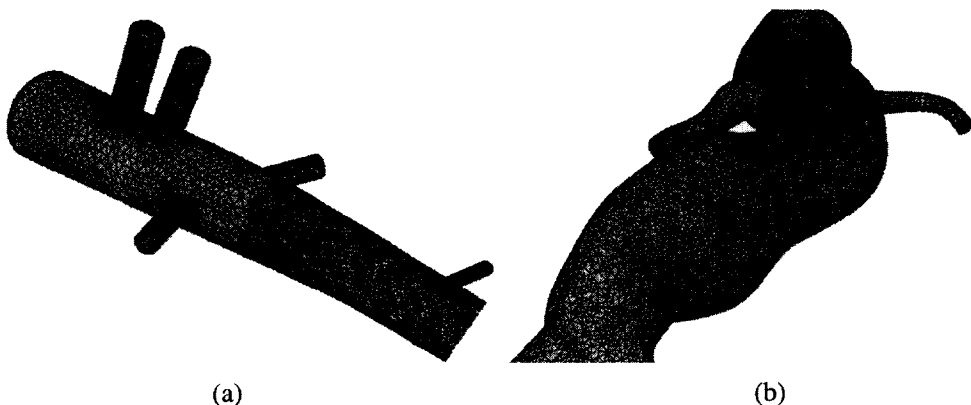


Fig. 13. Close-up views of finite element meshes of: (a) an idealized abdominal aorta model; (b) a patient-specific abdominal aorta model.

of thousands of nodal points. Robust scientific visualization software incorporating multiple techniques for interpreting and interrogating computational data sets is essential. The SpectrumTM Visualizer was utilized in the present work. Some of the visualization techniques that have been particularly effective in computational blood flow investigations are described below.

Consider the case of visualizing a vector field such as velocity. Multiple techniques including contour maps, isosurfaces, streamlines, particular traces and vector slices provide different insight into the results. For example, consider the flow in a model of the abdominal aorta. Fig. 14 displays four methods used to visualize the velocity vector field in the abdominal aorta at mid-diastole and examine the complex flow patterns downstream of the celiac, superior mesenteric and renal arteries. Note that one-half of the model has been removed for visualization purposes. Fig. 14(a) illustrates the use of a vector slice along the mid-sagittal plane. It is noted that the vector field reverses in direction along the posterior wall of the infrarenal abdominal aorta. Fig. 14(b) illustrates the use of a set of streamlines originating at the level of the diaphragm to visualize the velocity field. Swirling patterns in the velocity field can be discerned below the renal branches. Fig. 14(c) illustrates the use of particles to visualize the velocity field. A concentration of particles below the renal arteries is evident from this figure. Finally, Fig. 14(d) illustrates the use of an isosurface of velocity magnitude of 5 cm/s to identify the slow flow regions in the infrarenal aorta. It is noted that this surface pulls away from the wall in the infrarenal aorta and that further a vortex forms as evidenced by the second surface appearing along the posterior wall at the level of the renal arteries.

In addition to the visualization of qualitative features of the hemodynamic conditions, quantitative data must be extracted from the computed flow field. The four methods for describing the velocity field depicted in Fig. 14 are all computed directly from the velocity field and do contain quantitative information, but other techniques are more suitable for extraction of simpler quantitative measures of hemodynamic conditions. For example, to derive a quantitative measure of particle residence time one method is to define a region bounded by two planes,

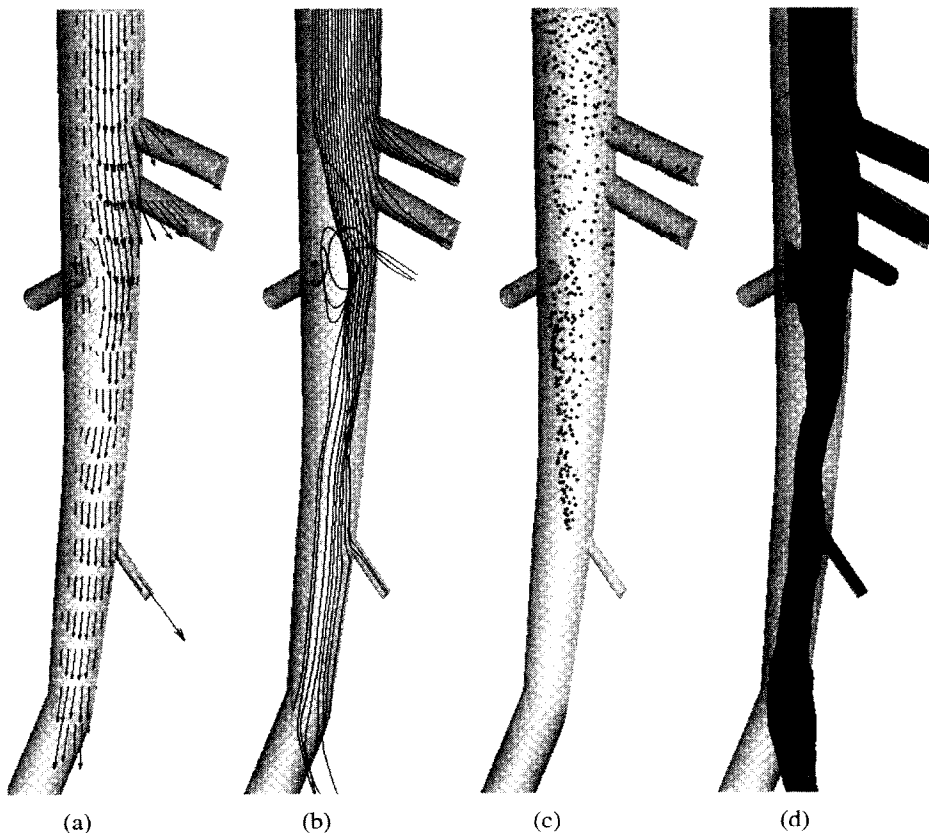


Fig. 14. Four techniques to visualize flow recirculation region at peak systole in idealized abdominal aorta model: (a) velocity vector field; (b) streamlines; (c) particles; (d) isosurface of velocity magnitude.

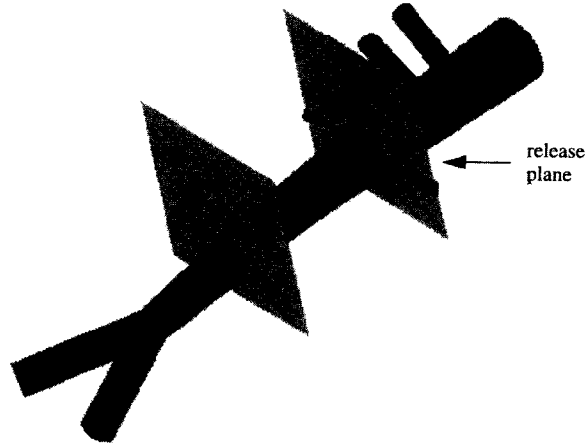


Fig. 15. Method to quantify particle residence time in idealized aorta model showing release plane and distal plane defining extent of region of interest.

as shown in Fig. 15, release particles from the proximal plane at a time zero and then determine the elapsed time until a given percentage of particles leaves the region bounded by the planes. A single number, the particle residence time, provides quantitative information regarding the extent of stasis in the flow field. Other measures of quantifying particle residence time are possible and might be more appropriate for determining spatial distributions of particles.

In addition to quantitative measures of particle residence time, quantitative values of the magnitude of the surface tractions are also of interest. The traction vector can be computed from the stress tensor, σ , and surface normal vector, $\mathbf{n}$, by $\mathbf{t} = \sigma \mathbf{n}$ and then the surface traction, $\mathbf{t}_s$, defined as the tangential component of the traction vector, can be computed from

$$\mathbf{t}_s = (\sigma - \sigma_{nn} \mathbf{I}) \mathbf{n} \quad (16)$$

where $\sigma_{nn} = \mathbf{n} \cdot \sigma \mathbf{n}$, and $\mathbf{I}$ is the identity tensor. Note that in the special case of the steady, axial flow of a Newtonian fluid in a circular cylinder, the magnitude of the surface traction reduces to the wall shear stress often quoted in the literature, $\tau_w = 4\mu Q / \pi r^3$, where μ is the viscosity, Q is the flow rate, and r is the lumen radius. However, due to the complex flow fields in the vascular system, the surface traction is a more general, and useful, measure of surface forces.

4. Validation and application of method

This section describes the application of the modeling and analysis methods developed to problems in blood flow. In order to assess the accuracy of the finite element method employed in the simplest possible setting, the pulsatile flow in a circular artery is computed and compared with the analytic Womersley solution. The second example presented, steady flow in an end to side graft-artery anastomosis, demonstrates the application of the methodology to a clinically relevant problem with moderate geometric complexity. The predicted velocity field is compared with published experimental data. The third example is pulsatile flow in the end-to-side graft-artery anastomosis considered in the previous example. The computed pulsatile flow results are again compared with published experimental data. The fourth example, pulsatile flow in a carotid bifurcation, is an example of diverging flow in an arterial branch prone to vascular disease. This example provides further evidence of the accuracy of the finite element solutions obtained by comparing the computed velocity profiles with published experimental data.

The fifth example, pulsatile flow in an idealized model of an abdominal aorta under simulated resting and exercise conditions, demonstrates the capabilities developed for creating complex geometric models, discretizing these models, solving the equations governing blood flow on parallel computers, and extracting and interpreting

the resultant data. The final example presented, an aneurysmal abdominal aorta flow model constructed from an individual patient's spiral CT data, illustrates the application of the methodology to complex anatomic and physiologic problems. The pre-operative model is modified to reflect a corrective surgical procedure demonstrating the potential application of this work to surgical planning.

4.1. Long, straight artery

The simplest example of pulsatile flow is provided by the flow of a Newtonian fluid in a rigid, long, straight tube with periodic velocity or pressure boundary conditions. An analytic solution, the Womersley solution described in Section 3, exists for this problem and can be used to test numerical methods.

A circular artery with a length of 3 cm and radius of 0.2 cm is subject to inflow velocity boundary conditions which are uniform in space and periodic in time. The time variation is described by a sinusoidal function $V(t) = \bar{V}(1 + \sin(2\pi t/T))$ with mean velocity, $\bar{V} = 13.5$ cm/s, and period, T , of 0.2 s. The fluid has a kinematic viscosity of 0.04 poise resulting in a mean Reynold's number of 135 and Womersley number of 5.6. Axi-symmetry was exploited and a quarter model constructed. This model was discretized into 51 568 tetrahedral elements and 11 325 nodes.

As seen in Fig. 16, the flow fully develops in a length equivalent to approximately ten diameters from the inlet as evidenced by axial velocity contours which do not vary with axial position thereafter. It should be noted that the model and results were reflected about the axis of symmetry for display purposes.

The Womersley solution for fully developed pulsatile flow, valid at the outlet of the artery, is compared with the computed axial velocity results at that position. First, the outflow volume is computed and compared to the inflow volume to verify that incompressibility is exactly satisfied. As observed in Fig. 17, the outflow volume flow rate does indeed match the inflow volume flow rate.

As observed in Fig. 18, the finite element solution resolves the theoretical velocity profile well with differences observed primarily along the centerline of the vessel in the second half of the cardiac cycle where the numerical results underpredict the exact solution. It is worth reiterating that the volume flow is exactly the same for the analytical and numerical solutions despite appearances to the contrary in Fig. 18. This apparent contradiction is easily resolved by recalling that, due to the axisymmetry of this problem, the undershoots in velocity near the axis can be countered by small overshoots near the edges. It should be noted that the level of discretization used for this example resulted in approximately 15 nodal points across the diameter, and a maximum error of less than 15%. As the exact solution is known for this problem, it should be possible to

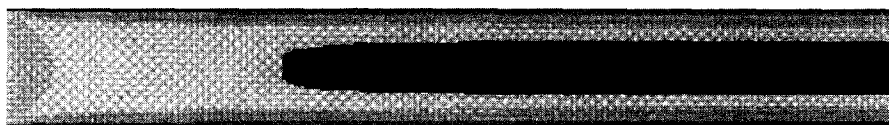


Fig. 16. Instantaneous axial velocity contours along the symmetry plane of a long, straight artery.

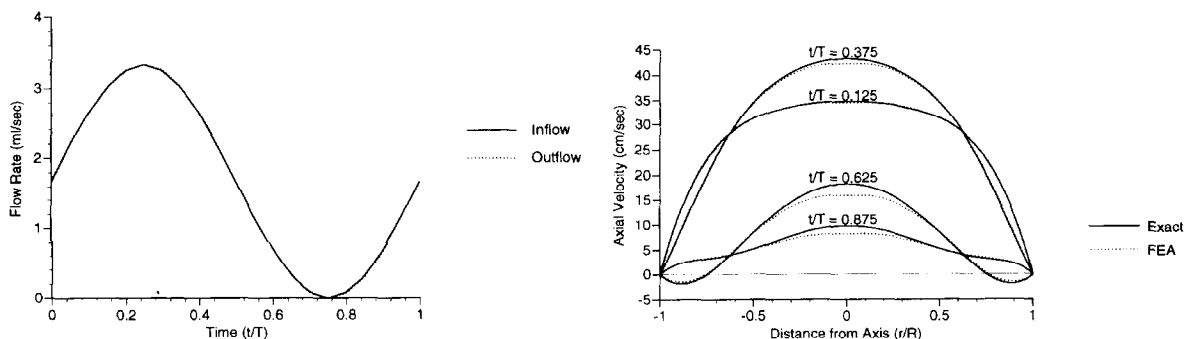


Fig. 17. Comparison between inflow and outflow volume flow rate for a long, straight artery.

Fig. 18. Computed and theoretical velocity profiles at various times in the cardiac cycle for the flow in a long, straight artery.

further improve the agreement between the finite element results and the Womersley solution with a finer mesh, but it is at present unclear what level of refinement would be required.

4.2. Steady flow in an end-to-side anastomosis

As noted in Section 1, the construction of an end-to-side vascular bypass graft is frequently employed by surgeons to bypass a diseased or injured segment of an artery. Hemodynamic conditions, including low shear stress, are hypothesized to be one of the factors in the localization of the intimal thickening observed at the toe and floor of the anastomosis [2]. Computational simulations of blood flow in graft-artery anastomoses can provide a means to quantify the hemodynamic conditions for comparison with observed sites of intimal thickening. Further, once the mechanisms of intimal thickening are better understood, the analysis procedures advocated herein can be used to design anastomoses which minimize adverse hemodynamic conditions.

Loth [29] describes the construction of an in vitro end-to-side graft model and velocity fields measured with LDA under a variety of steady and pulsatile flow conditions. Using fluid dynamic similarity principles, the data from the in vitro model was scaled to match an in vivo canine ilio-femoral anastomoses model. The model geometry for the present computational study was constructed so as to correspond to the dimensions of the in vivo model presented by Loth [29] and is depicted in Fig. 19.

The geometric solid model created in the Virtual Vascular Laboratory is shown in Fig. 20. This solid model corresponds to a high degree of accuracy with the model dimensions meticulously detailed by Loth [29]. A finite element mesh, shown in Fig. 21, was generated with 46 453 nodes and 246 286 linear tetrahedral elements. The maximum element size was specified to be 1/15th of the host artery diameter.

The flow conditions were specified to correspond to the steady flow experiments of Loth [29] scaled to in vivo conditions. Namely, a graft input flow rate of 1.9 ml/s, viscosity of 0.035 g/s/cm, and density of 1.05 g/ml. These parameters yield a Reynolds number of 208, based on the graft flow rate and host artery

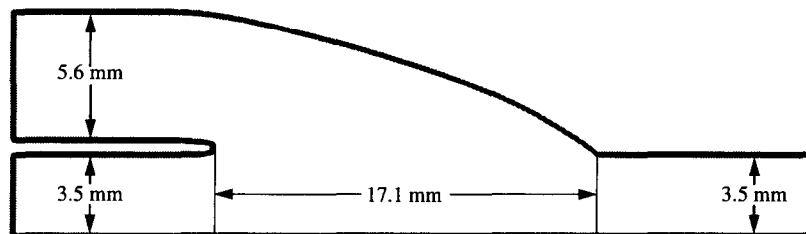


Fig. 19. Selected dimensions of the end-to-side anastomosis model.

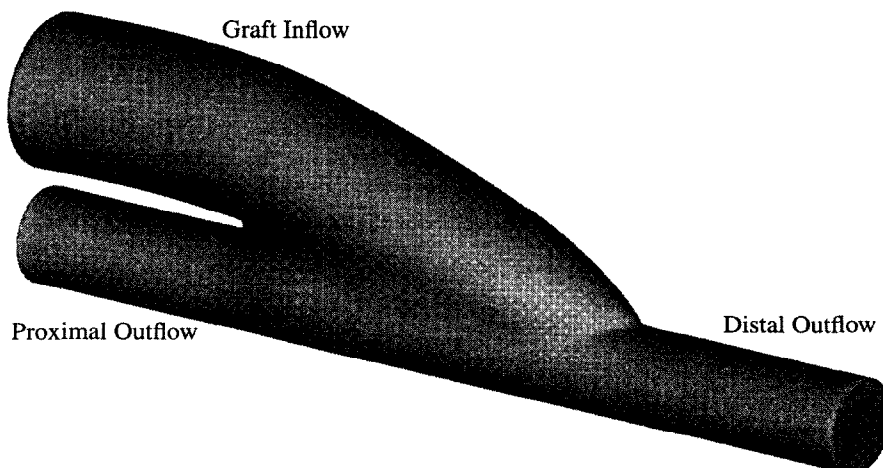


Fig. 20. Geometric solid model of the end-to-side anastomosis model.

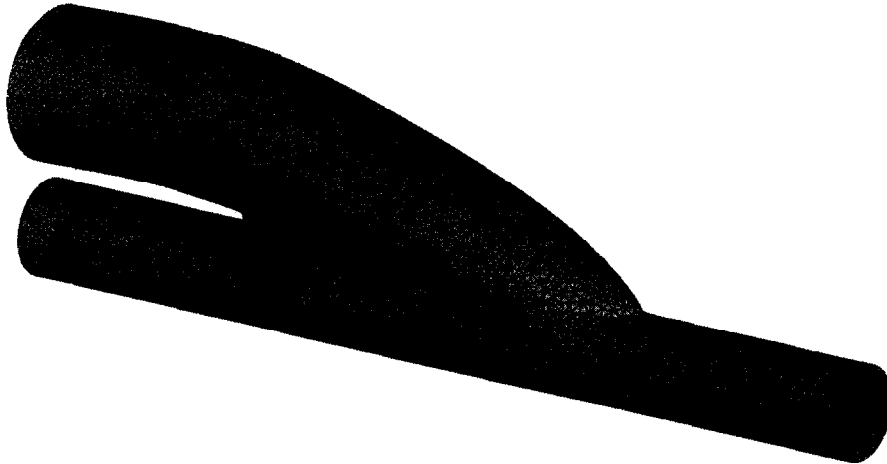


Fig. 21. Finite element mesh for the end-to-side anastomosis model.

diameter. A parabolic profile was specified at the graft inlet to achieve the desired input flow rate of 1.9 ml/s. The division of flow between the proximal and distal outlet segments was set at 20:80 by prescribing a parabolic outflow velocity profile at the proximal outlet segment and setting the pressure at the distal outlet segment to zero.

The velocity vector field along the symmetry plane is depicted in Fig. 22. The length of the arrows corresponds to the magnitude of velocity. The darker colors of arrows along the center line also denote a higher velocity magnitude. It is observed that, due to the proximal outflow, a region of flow recirculation does not develop at this Reynold's number of 208. Further, the reduction in flow velocity as the blood enters the anastomotic region is observed, as is the increase in velocity as the blood enters the distal outflow segment.

Fig. 23 shows the comparison between the computed velocity profiles and experimental data from Loth [29] for the x and y components of velocity along lines through the midplane in the plane of the anastomosis. It is observed that the computational velocity results are within 10% of the experimental values at most locations, and that the qualitative comparison is also favorable.

Fig. 24 shows the comparison between the computed velocity profiles and experimental data from Loth [29] for the x and y components of velocity along lines transverse to the plane of the anastomosis. Again, the comparison between computational and experimental data is favorable.

Fig. 25 displays results for the pressure drop and shear stress variation, denoted by $|\tau_w|$, along the floor of the host artery from the proximal outflow segment to the distal outflow segment. Two points are worth noting. First, the shear stress magnitude is less than 5 dynes/cm² for most of the artery floor in the anastomotic region, while the pressure difference reaches a maximum of 400 dynes/cm² in the anastomotic region. Second, the magnitude of the variation in pressure along the length of the host artery is an order of magnitude greater than the shear stress.

Fig. 26 shows the magnitude of the surface traction force vector along the surfaces of the entire end-to-side anastomosis model. The low shear region (dark blue) is observed along the hood and the floor of the artery and a high shear region (red) is observed at the distal outflow segment. Note that the two bands of elevated shear along the side walls of the anastomotic region are artifacts of the finite element discretization.

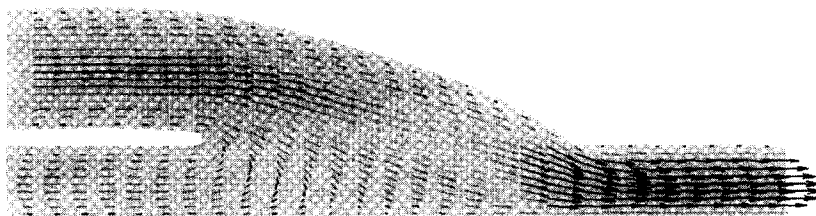


Fig. 22. Velocity vector field along the symmetry plane for steady flow in an end-to-side anastomosis.

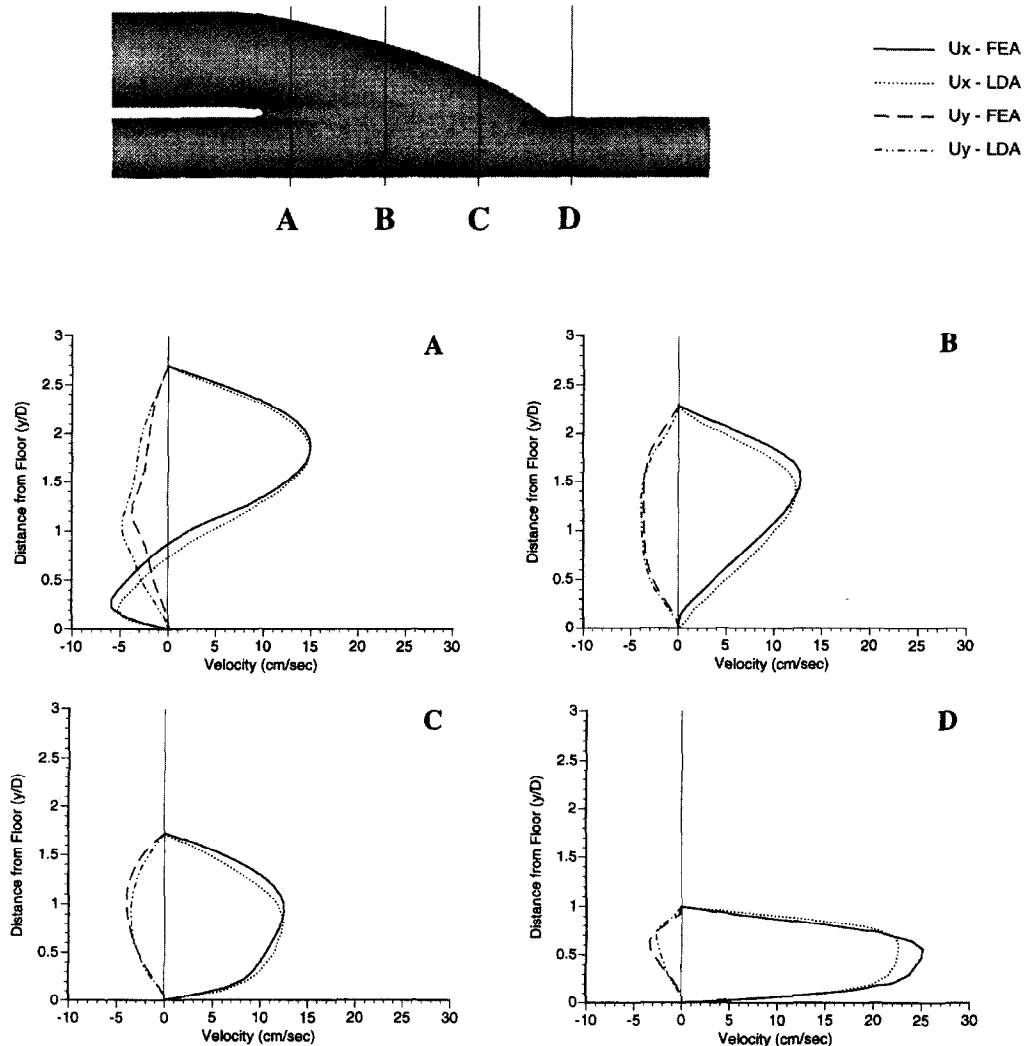


Fig. 23. Velocity profiles in the symmetry plane for steady flow in an end-to-side anastomosis.

4.3. Pulsatile flow in an end-to-side anastomosis

The geometric model was described in the previous example, and is shown in Fig. 20. For this example, a symmetry condition was utilized to reduce computing requirements, so only one half of the model was discretized into 21 150 nodes and 103 800 tetrahedral elements. The finite element mesh is shown in Fig. 27.

The flow conditions were specified to correspond to the pulsatile flow experiments of Loth [29] scaled to in vivo conditions. A Womersley profile was specified at the graft inlet to achieve the desired input flow waveform depicted in Fig. 28. The division of flow between the proximal and distal outlet segments was again set at 20:80 by prescribing a Womersley outflow velocity profile at the proximal outlet segment (POS) and setting the pressure at the distal outlet segment (DOS) to zero.

The velocity vector field along the symmetry plane at peak systole (phase of 45°) and mid-diastole (phase of 240°) is depicted in Fig. 29. The flow patterns under pulsatile flow conditions are far more complex than those observed during steady flow conditions (Fig. 22). Vortices form at different locations in the anastomotic region and are convected and dissipated during the cardiac cycle. Note in particular the regions of flow recirculation in the anastomotic region at mid-diastole.

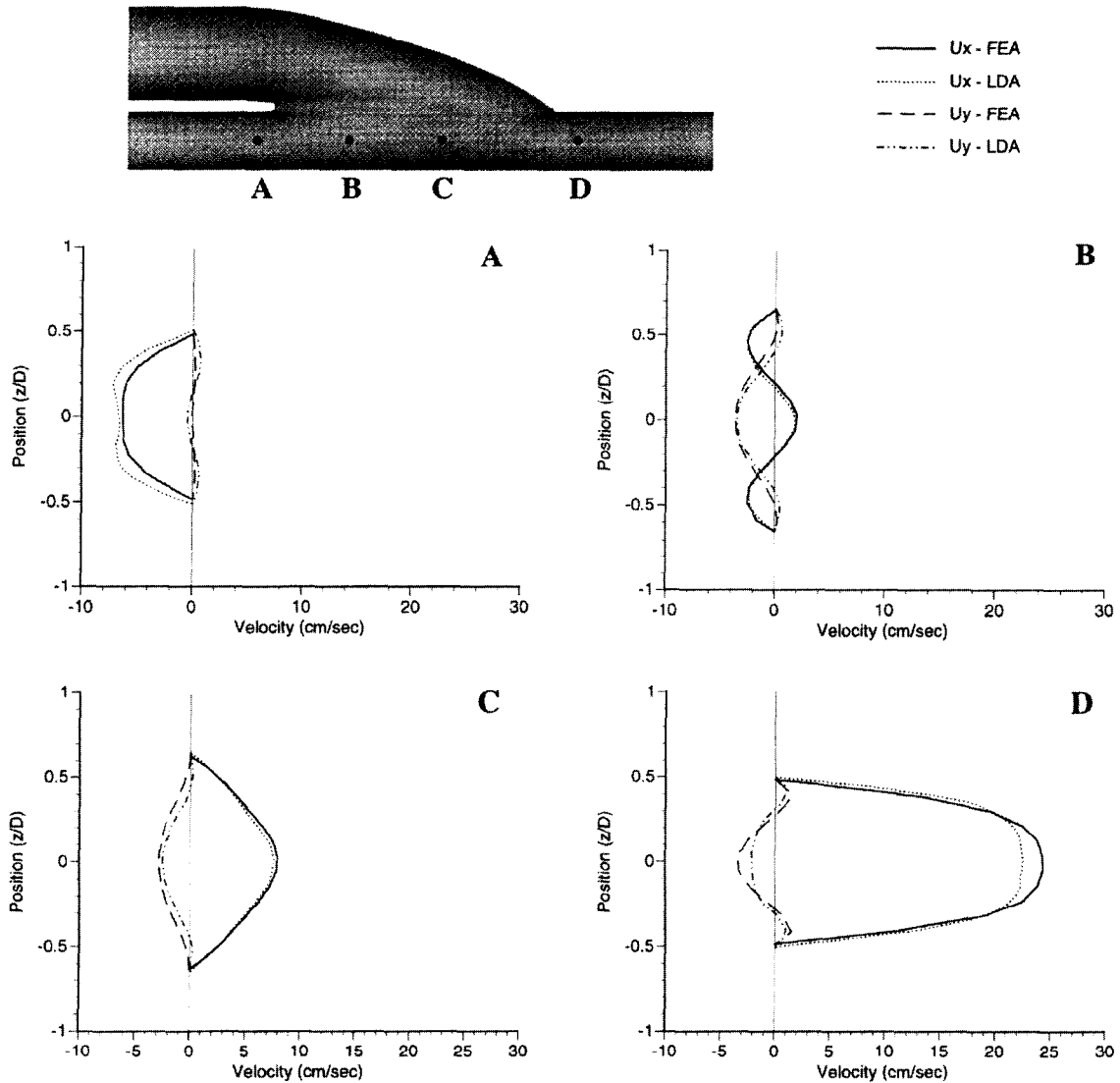


Fig. 24. Velocity profiles transverse to the symmetry plane for steady flow in an end-to-side anastomosis.

Fig. 30 shows the comparison between the computed velocity profiles and experimental data from Loth [29] for the x components of velocity along a line through the midplane in the plane of the anastomosis near the proximal end of the anastomotic region. It is observed that the computational results compare favorably qualitatively and quantitatively with the experimental data (less than 10% error at most points).

Fig. 31 shows the comparison between the computed velocity profiles and experimental data from Loth [29] for the y components of velocity along a line through the midplane in the plane of the anastomosis near the proximal end of the anastomotic region. Again, the comparison between computational and experimental data is satisfactory.

Fig. 32 shows the comparison between the computed velocity profiles and experimental data from Loth [29] for the x components of velocity along a line through the midplane in the plane of the anastomosis near the center of the anastomotic region. The comparison between computational and experimental data at this location is very good for all times in the cardiac cycle.

Fig. 33 shows the comparison between the computed profiles and experimental data for the y components of velocity along a line through the midplane in the plane of the anastomosis near the center of the anastomotic

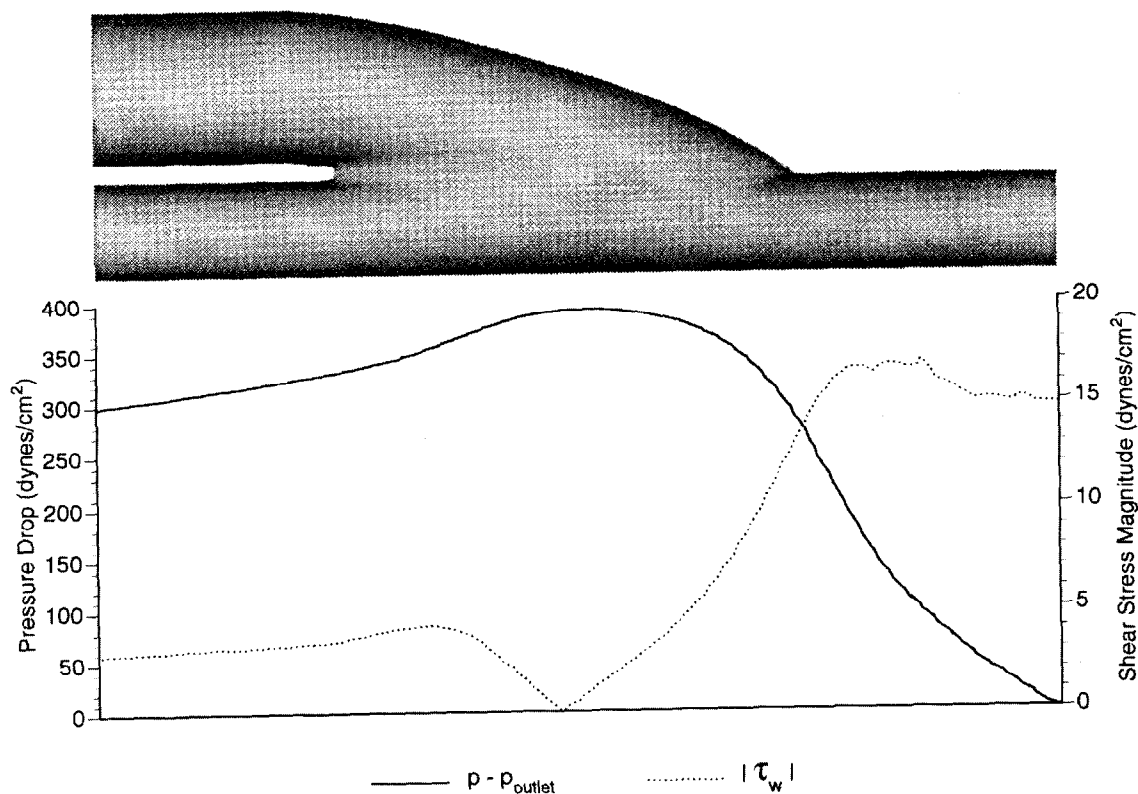


Fig. 25. Pressure and shear along the host artery floor for steady flow in an end-to-side anastomosis.

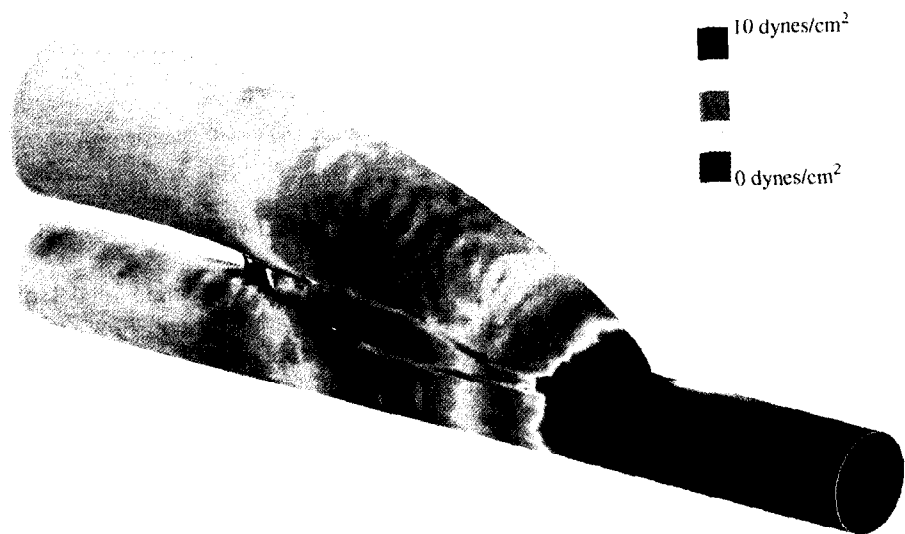


Fig. 26. Magnitude of surface traction for steady flow in an end-to-side anastomosis model.

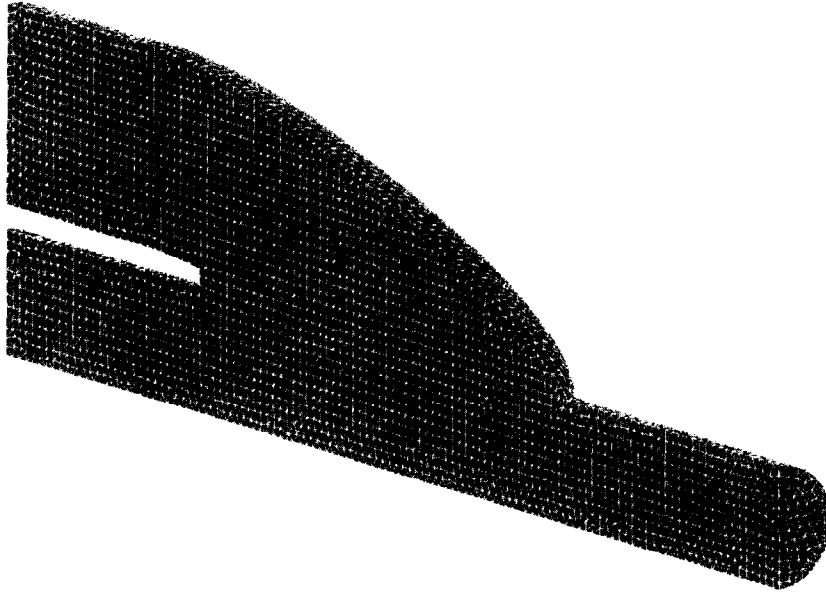


Fig. 27. Finite element mesh for pulsatile flow in an end-to-side anastomosis.

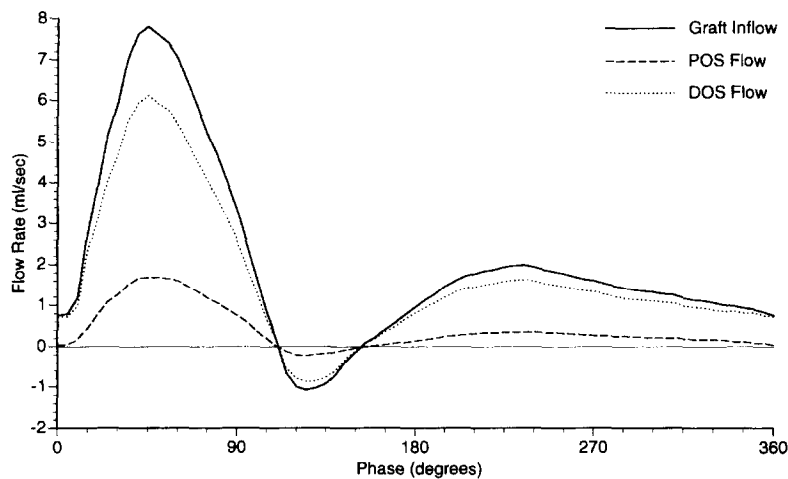


Fig. 28. Volume flow waveforms for pulsatile flow in an end-to-side anastomosis.

region. Again good qualitative and quantitative agreement is obtained between computational and experimental data at this location for all times in the cardiac cycle.

This example demonstrates that the pulsatile flow in an anatomically accurate model of an end-to-side artery-graft anastomosis can be computed accurately, given known geometry and flow conditions. Thus, the computational methods developed have the potential to be a useful tool for the design of graft configurations which minimize energy dissipation, flow recirculation, and flow stasis.

4.4. Carotid bifurcation

A problem of considerable clinical importance is the pulsatile flow in a carotid artery. As noted previously, plaques localize in the carotid sinus region where wall shear stress is low and particle residence time is high [43].

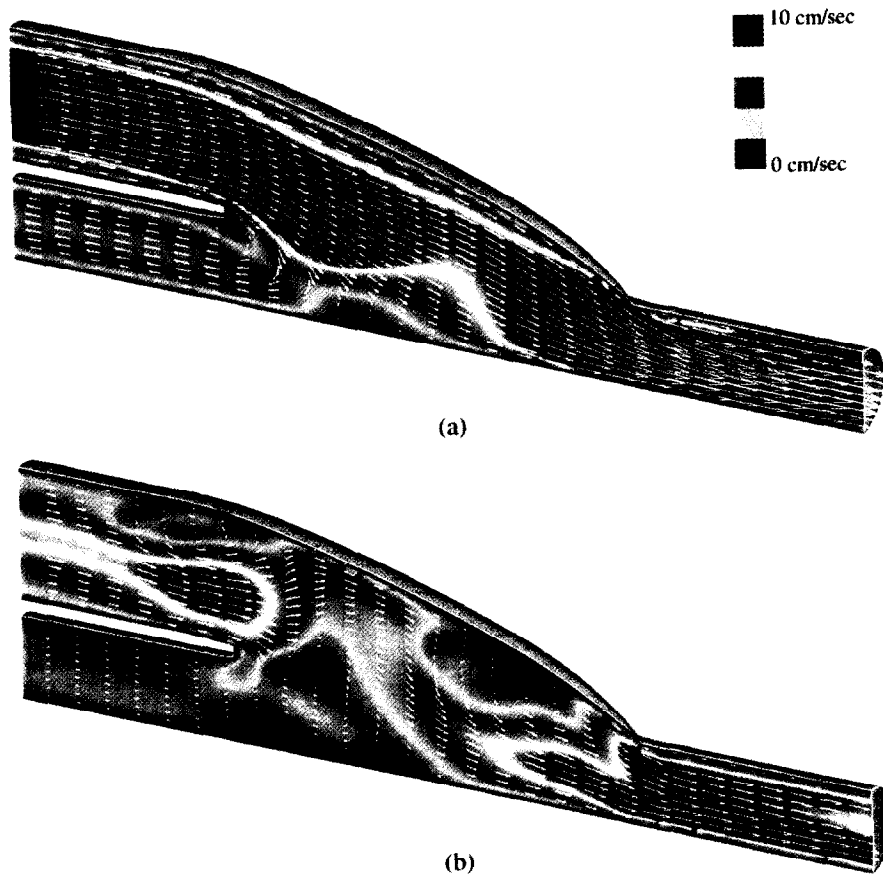


Fig. 29. Velocity contours and vector field at: (a) peak systole; (b) mid-diastole for the pulsatile flow in an end-to-side anastomosis.

The model geometry was constructed using the *in vivo* dimensions detailed in [25]. The model geometry is shown with dimensions in Fig. 34. A finite element mesh with 34 513 nodes and 178, 230 tetrahedral elements was constructed using the dimensions detailed in [2]. The symmetry of the model was exploited to reduce computational requirements. Fig. 35 shows a close-up view of the finite element mesh in the region of the bifurcation.

Pulsatile flow conditions, shown in Fig. 36, were specified by using the flow rates given in [3]. Fully developed pulsatile flow boundary conditions, employing the Womersley solution, were specified for the inflow velocity in the common carotid artery and the outflow velocity for the external carotid artery. A zero pressure boundary condition was specified for the outflow of the internal carotid artery.

The analysis was performed using a time-marching algorithm for three cardiac cycles with 200 time steps per cycle on a single processor HP 755/125 workstation. Fig. 37 depicts the velocity vector field at peak systole (phase of 35°) and mid diastole (phase of 240°) in the symmetry plane of the carotid bifurcation. The length of the vectors has been normalized to show the flow reversal along the outer wall of the carotid sinus region; the colors correspond to the magnitude of the velocity with blue being lowest and red highest. It is noted that a vortex forms and the velocity vectors reverse in direction near the wall of the carotid sinus at peak systole. This flow reversal disappears during diastole, but the magnitude of velocity is still low along the outer wall of the carotid sinus.

Fig. 38(a) and (b) depict isosurfaces of velocity magnitude equal to 0 cm/s (blue), 5 cm/s (light blue), 10 cm/s (green), 15 cm/s (yellow), and 20 cm/s (red), at times just after peak-systole and mid-diastole, respectively. Half of the model is removed for visualization purposes. The maximum value of velocity magnitude is approximately 600 cm/s in the carotid artery, so an isosurface of 5 cm/s represents relatively slow blood flow. Note that in the carotid sinus region there is a large separation between the zero velocity isosurface

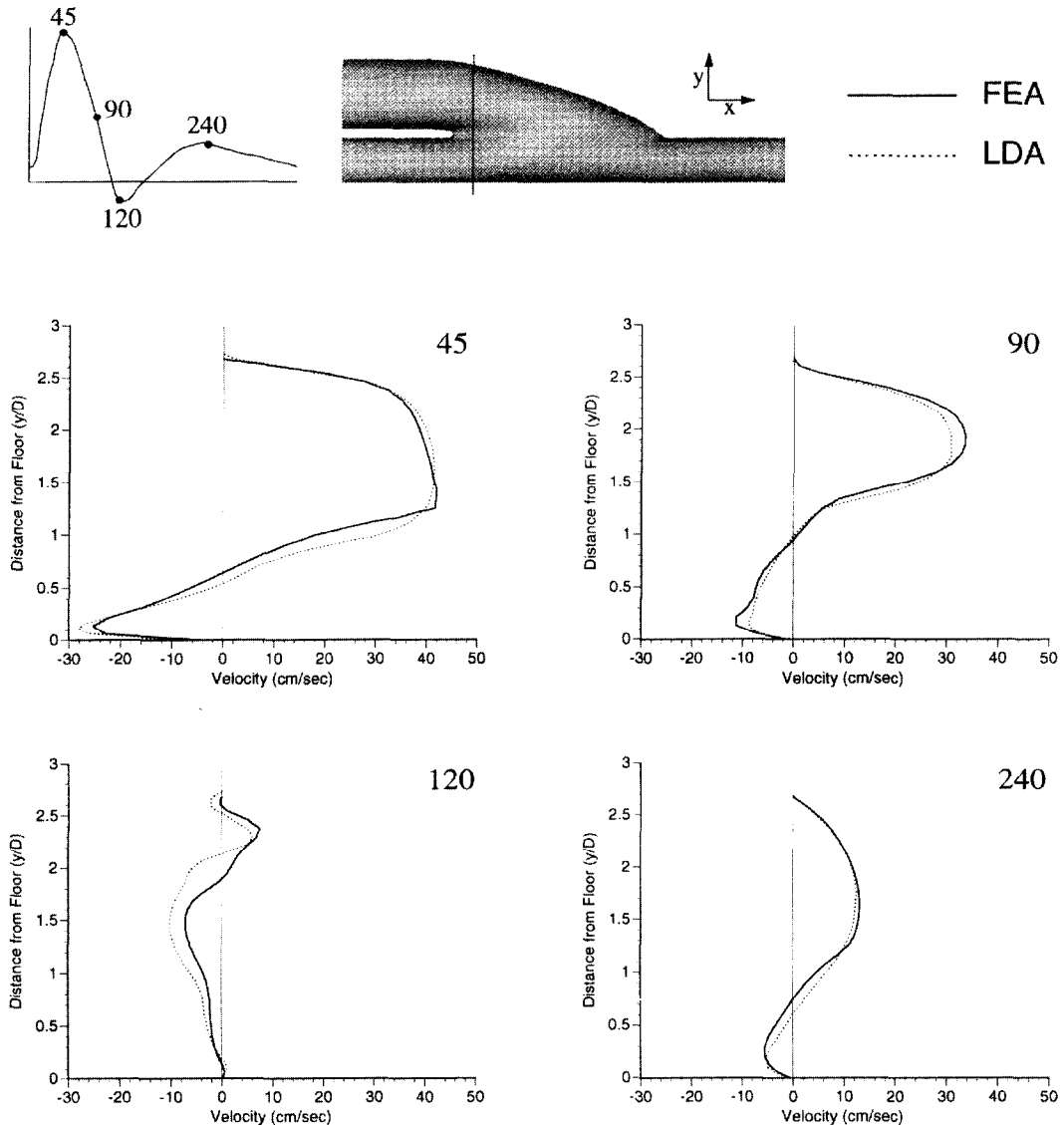


Fig. 30. Computed and experimental velocity component in x direction at a cross-section near the proximal end of the anastomotic region.

representing the wall and the 5 cm/s isosurface. This gap corresponds to a low velocity region and is observed at all times during the cardiac cycle.

A region of slow blood flow exists along the outer wall of the carotid sinus in precisely the region where plaques localize as observed in sections B and C in Fig. 1(a). Also, note the closeness of the isosurfaces along the inner wall of the bifurcation near the flow divider, as well as in the regions devoid of plaque in Fig. 1(a), sections A and D. This corresponds to steep gradients of velocity and high shear stress in these regions. This analysis supports the hypothesis that plaques localize in the carotid bifurcation in regions of low wall shear stress and low blood velocity.

4.5. Idealized abdominal aorta

The aorta is the largest artery in the human body and performs the critical function of transferring oxygenated blood from the left ventricle of the heart to the arteries. The abdominal aorta is the segment of the aorta from the level of the diaphragm to the aortic bifurcation just below the navel. Clinically, it is observed that atherosclerotic

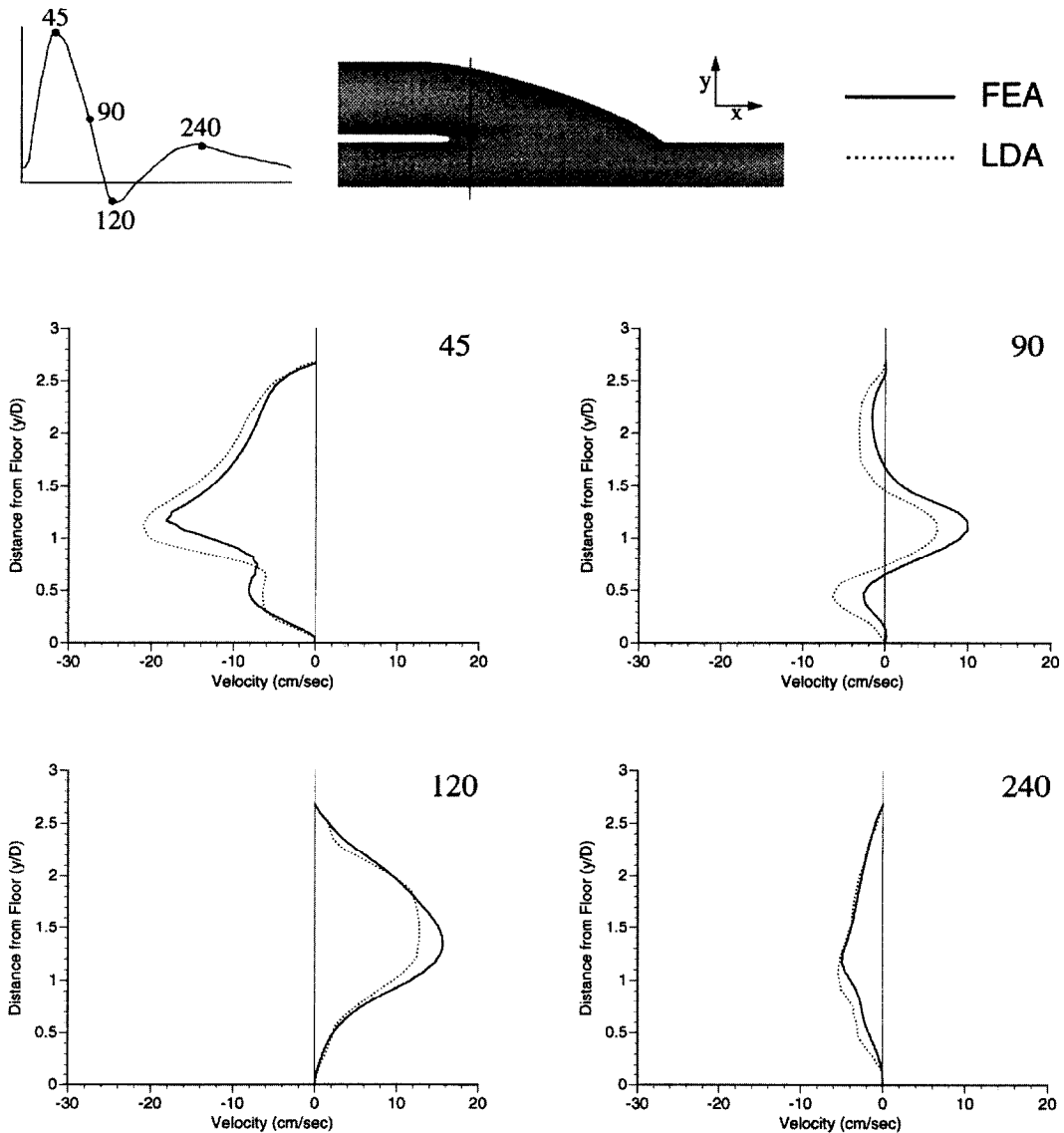


Fig. 31. Computed and experimental velocity component in y direction at a cross-section near the proximal end of the anastomotic region.

disease develops first in the abdominal aorta, and is much more common in the abdominal aorta (i.e. below the diaphragm) than the segment of the aorta above the diaphragm, the thoracic aorta. It has been hypothesized that this is due in part to 'adverse hemodynamic conditions' such as low shear stress and high particle residence time resulting from the complex flow patterns in the abdominal aorta [46]. The pulsatile flow in the abdominal aorta is particularly complex under normal resting conditions as a result of the multiple branches at the level of the diaphragm which extract blood for the organs in the abdomen.

Fig. 39 displays the computer model constructed to investigate blood flow in the abdominal aorta under simulated normal resting conditions and various levels of simulated lower limb exercise. The anatomical dimensions and flow conditions were obtained from Moore et al. [31]. The surface of a finite element mesh with 268 563 tetrahedral elements and 58,151 nodes is displayed in Fig. 40. Note the refinement of the mesh in the region below the renal arteries. The flow computations were performed on a parallel computer, requiring the computational domain to be divided into subdomains. The result of the domain decomposition for 16 subdomains is shown in Fig. 41. The nonlinear evolution equations were solved over 3 cardiac cycles with 200 time steps per cardiac cycle on an IBM SP-2 parallel computer.

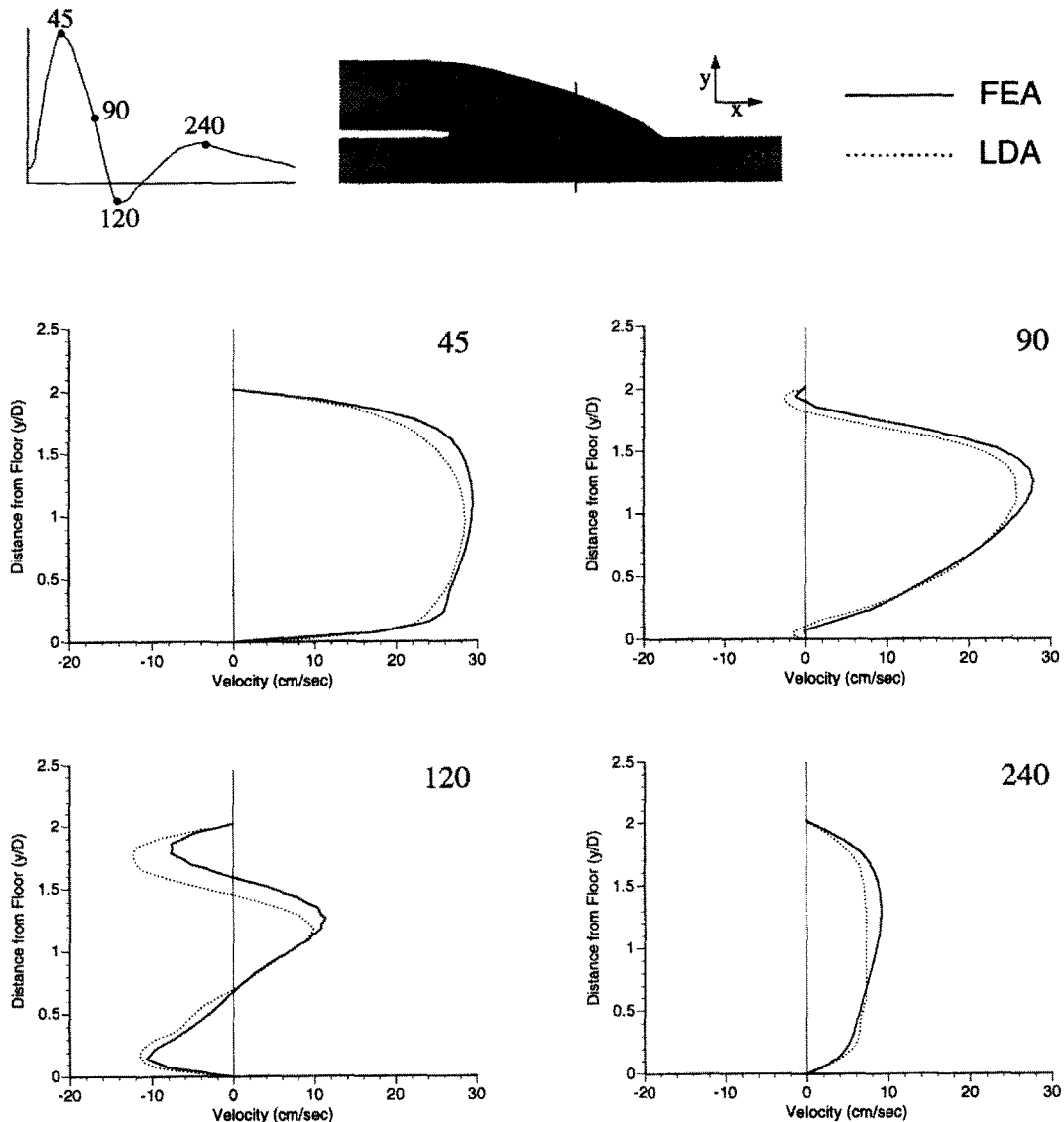


Fig. 32. Computed and experimental velocity component in x direction at a cross-section near the center of the anastomotic region.

Under resting conditions, approximately 70% of the blood that enters the abdominal aorta is extracted by the celiac, superior mesenteric, and renal arteries. The majority of the remaining 30% flows down the infrarenal segment through the bifurcation into the legs. Under conditions of moderate lower limb exercise the heart rate and cardiac output can easily double. Furthermore, much of the blood that enters the abdominal aorta is diverted to the legs from the abdominal organs. Consequently, the flow through the infrarenal segment, to the legs, can increase by a factor of 5 or more [46].

The flow rates as a function of time for the inflow and all branch vessels is shown in Fig. 42. The suprarenal and infrarenal flow rate functions were obtained from Moore [30], and the celiac, superior mesenteric, renal and inferior mesenteric functions were computed to yield the mean flow rates reported by Moore et al. [31].

A contour slice of axial velocity and a vector slice of the velocity field is displayed in Fig. 43 under simulated resting and exercise conditions at mid-diastole along the midplane of the aorta. Note that under resting conditions a large vortex develops along the posterior wall of the aorta. This region of flow stasis and high particle residence time disappears with moderate levels of simulated lower limb exercise, such as might be obtained with a brisk walk.

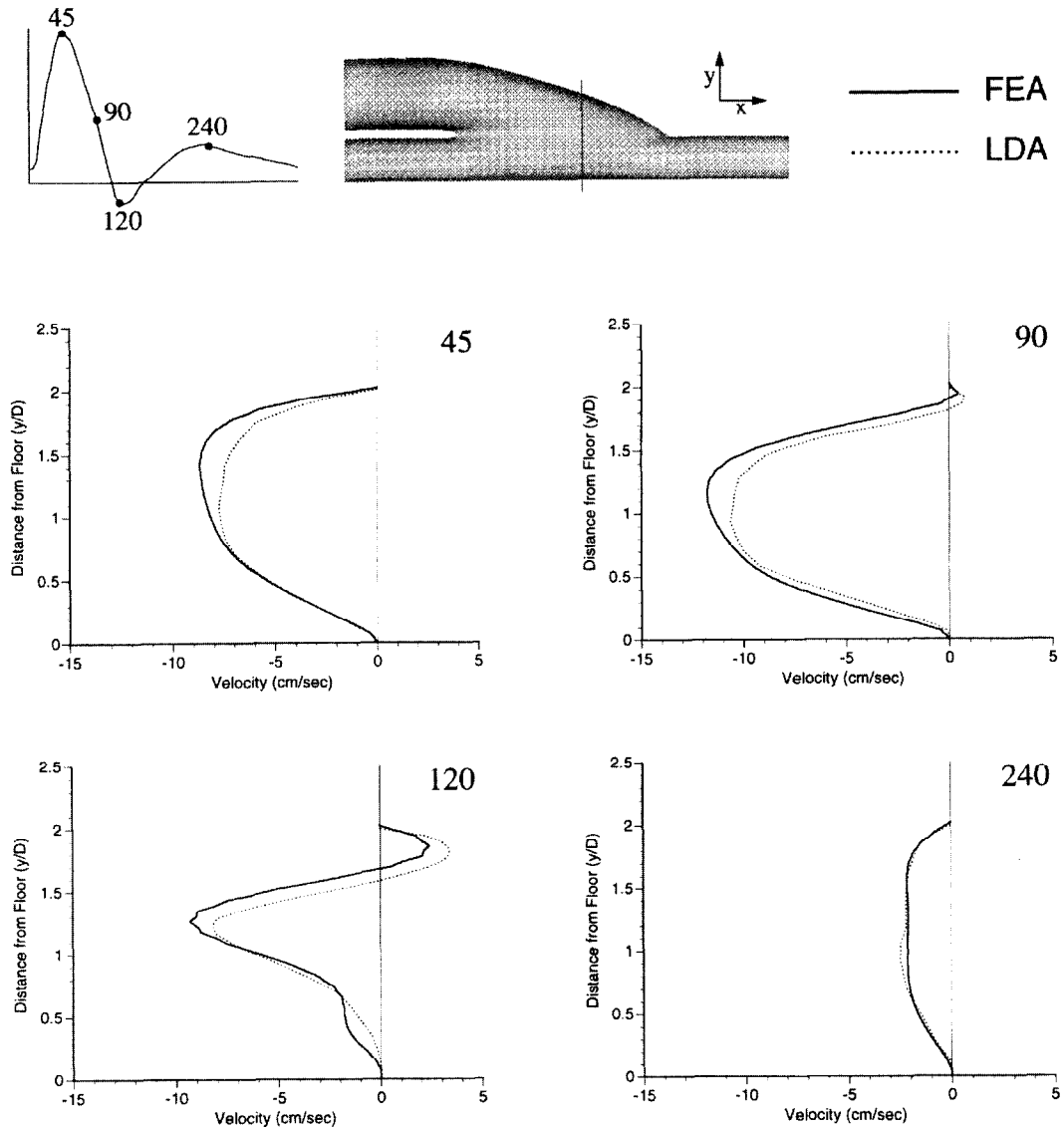


Fig. 33. Computed and experimental velocity component in y direction at a cross-section near the center of the anastomotic region.

Contour plots of time-averaged (mean) wall shear stress are displayed in Fig. 44 under simulated resting and exercise conditions. Note that under resting conditions a region of low wall shear stress develops below the renal artery branch along the lateral and posterior walls. Shear stresses are elevated with moderate levels of simulated lower limb exercise to a value in excess of 5 dynes/cm^2 in almost the entire abdominal aorta, above the level of the normally lesion-resistant suprarenal aorta under resting conditions. Wall shear stress is further elevated under vigorous exercise conditions to a level in excess of 10 dynes/cm^2 in almost the entire abdominal aorta.

Fig. 45 displays particles in the abdominal aorta under simulated resting and exercise conditions one second after release at the level of the diaphragm. Note that under resting conditions particles tend to accumulate below the renal arteries, while these particles are advected downstream under moderate and vigorous exercise conditions. In fact, under vigorous exercise conditions, most of the particles have left the abdominal aorta one second after release.

Particle residence time was quantified in the infrarenal abdominal aorta using the method described in Section 3 and shown in Fig. 15. Particles are released from the plane just distal to the renal arteries. The particle residence time is defined as the time required for 90% of released particles to clear the infrarenal aorta region

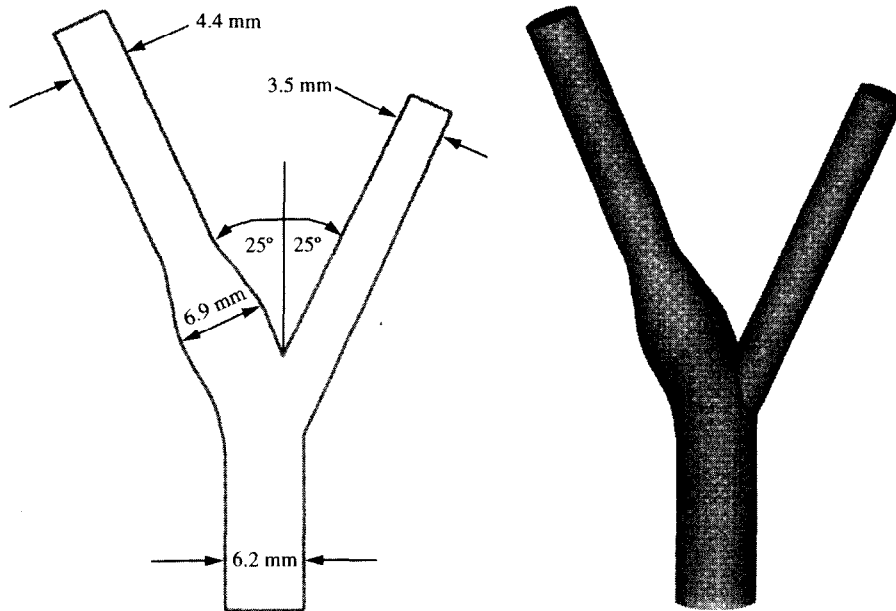


Fig. 34. Model dimensions and geometric solid model for carotid artery.



Fig. 35. Close-up of finite element mesh on the symmetry plane of the carotid artery.

bounded by the two planes. Under resting conditions, 3.3 s is required, whereas under moderate exercise conditions, only 1.0 s is required, and under vigorous exercise conditions, 0.6 s is required. While this measure of particle residence time does not provide information about locations where particle residence time is high, it is a useful measure of the gross effects of the flow on particles under different physiologic conditions and could be computed at multiple positions along the vessel.

4.6. Patient-specific abdominal aorta

4.6.1. Pre-operative analysis

The preceding examples demonstrate that computer models can be constructed and the blood flow characterized for idealized models representing the average anatomy of a group of individuals. These idealized

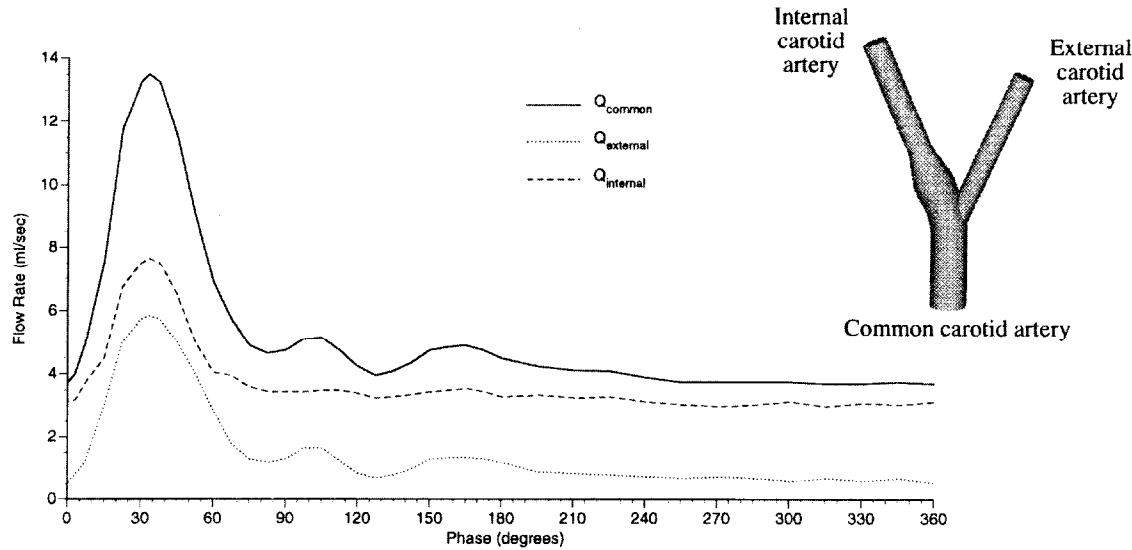


Fig. 36. Pulsatile flow conditions for the carotid artery simulation.

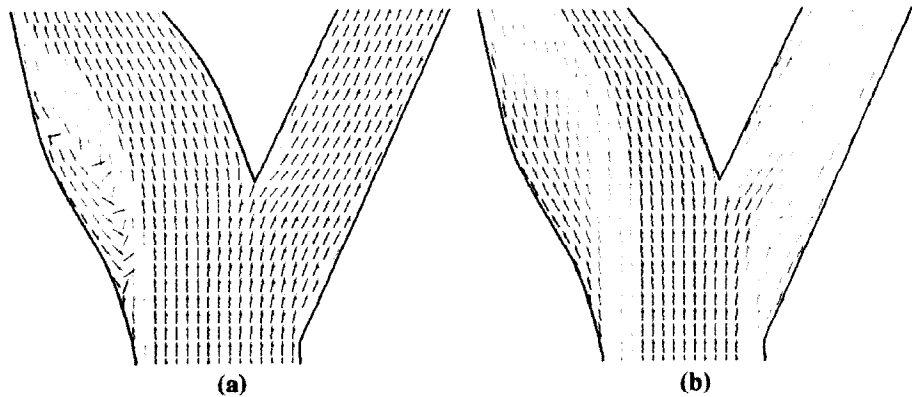


Fig. 37. Computed flow results for pulsatile flow in a carotid bifurcation. Normalized velocity vectors along the symmetry plane at: (a) peak systolic flow; (b) mid-diastolic flow. Range of velocity is from 0 to 20 cm/s with blue denoting low velocity and red denoting high velocity.

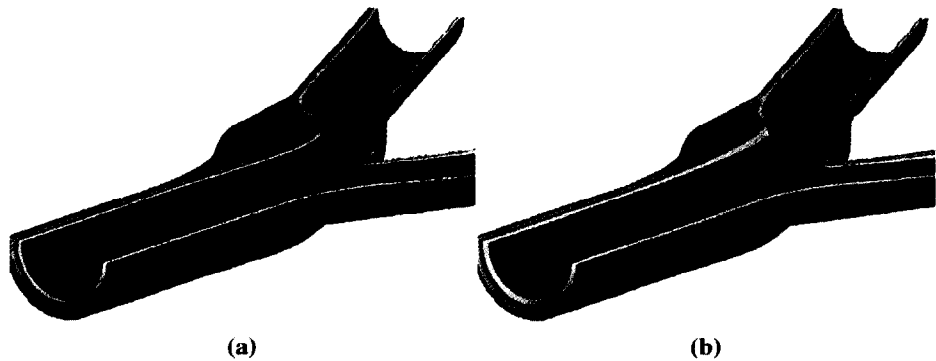


Fig. 38. Computed flow results for pulsatile flow in a carotid bifurcation. Isosurfaces of velocity magnitude at: (a) peak systolic flow; (b) mid-diastolic flow.

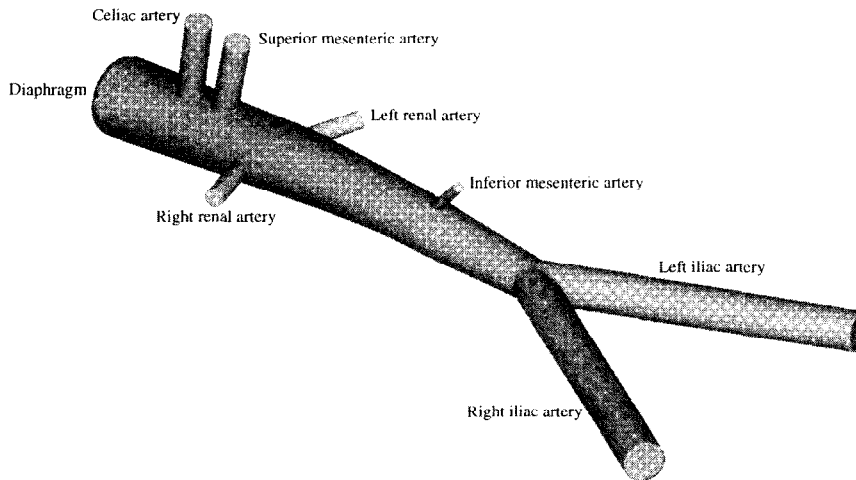


Fig. 39. Abdominal aorta model with branches identified.

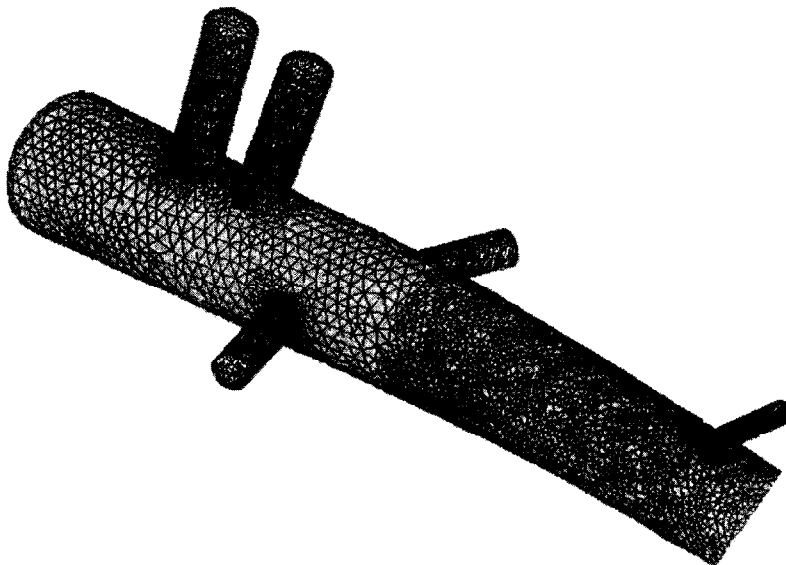


Fig. 40. Close-up of abdominal aorta finite element mesh.

models can be modified in a systematic fashion to investigate the effect of variations in anatomy on flow conditions and are useful for quantifying hemodynamic factors in blood vessels prone to vascular disease. However, to determine the exact flow conditions in a given individual's vascular system for clinical diagnoses or surgical planning, models which more faithfully represent individual anatomical features and flow conditions are necessary. These models are not created using CAD data, but instead by extracting anatomical information from medical imaging sources including computed tomography (CT) and magnetic resonance imaging (MRI). Fig. 46 displays a geometric model reconstructed from spiral CT data of a patient with an aneurysm in the abdominal aorta. The mural thrombus in the aneurysmal sac and calcifications along the vessel wall were identified in addition to the flow domain. A finite element mesh was created with 315 608 tetrahedral elements and 68 468 nodes and is shown in Fig. 46. The approximately 270 000 nonlinear equations governing the blood flow in this model were solved for three cardiac cycles with 200 time steps per cardiac cycle on an 8 processor SGI Power Challenge.

Fig. 47 depicts a slice of the axial (primary) component of velocity along a plane through the aortic bifurcation. A mesh slice plane was used to remove the top of the aneurysm for visualization of the flow. The

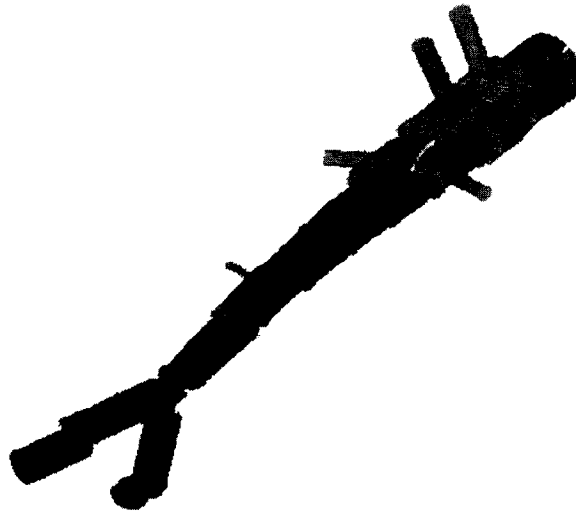


Fig. 41. Subdomains created for parallel computations in the abdominal aorta model.

top part of the model is then redisplayed in wireframe. Also shown is the large mural thrombus along the side wall of the aneurysm. As seen in Fig. 47, the flow decelerates as it enters the aneurysmal cavity and a vortex develops in the anterior-posterior direction as noted by the negative flow (blue) along the back wall of the aorta and the positive flow (red) along the anterior wall just above the bifurcation. Reverse and stagnant flow is observed close to the wall of the aneurysm along the anterior and posterior wall of the aorta. These numerical investigations into blood flow in an individual patient's vasculature can be used to augment the anatomical data provided by medical imaging diagnostics.

4.6.2. Post-operative analysis

The geometric model used for the pre-operative analysis was modified to reflect a surgical corrective procedure in which the aneurysm was isolated with the insertion of an intravascular graft (extending from below the renal arteries into the right common iliac artery), the left common iliac vessel was blocked, and a femoral-femoral bypass was created to restore flow to the left leg. To simulate the insertion of the stent graft, the abdominal aorta dimensions were reduced in the aneurysmal region. A portion of the left common iliac vessel was removed and boundary conditions for velocity were specified to be zero at the two exposed ends. A bypass graft was added between the right and left iliac femoral arteries. The outflow boundary conditions at the distal ends were maintained from the pre-surgical simulation to reflect the same distal resistances. Fig. 48 illustrates the new solid model used for the post-operative analyses.

Fig. 49 depicts the flow streamlines at peak systole for the post-operative simulation. The recirculation of flow in the occluded left common iliac artery and in the right femoral artery anastomotic region are examined more closely. The hemodynamic conditions in these regions are highly complex. There may be greater risk of thrombus generation and intimal thickening in these regions than in the high shear right common iliac artery.

This example illustrates that computational models can be created to reflect realistic anatomic configurations obtained from medical imaging data. Further, these models can be subsequently modified to pre-operatively examine surgical options and the effect of a possible procedure on hemodynamic conditions. Greater understanding of the role of hemodynamic conditions in vascular adaptation and disease localization will be required to determine long-term outcomes.

5. Conclusions

Hemodynamic conditions, including velocity, shear, and pressure, play an important role in the modulation of vascular adaptation and the localization of vascular disease. Consequently, understanding the local hemodynamic environment in a region of the vascular system is an important field of research. As we enter the

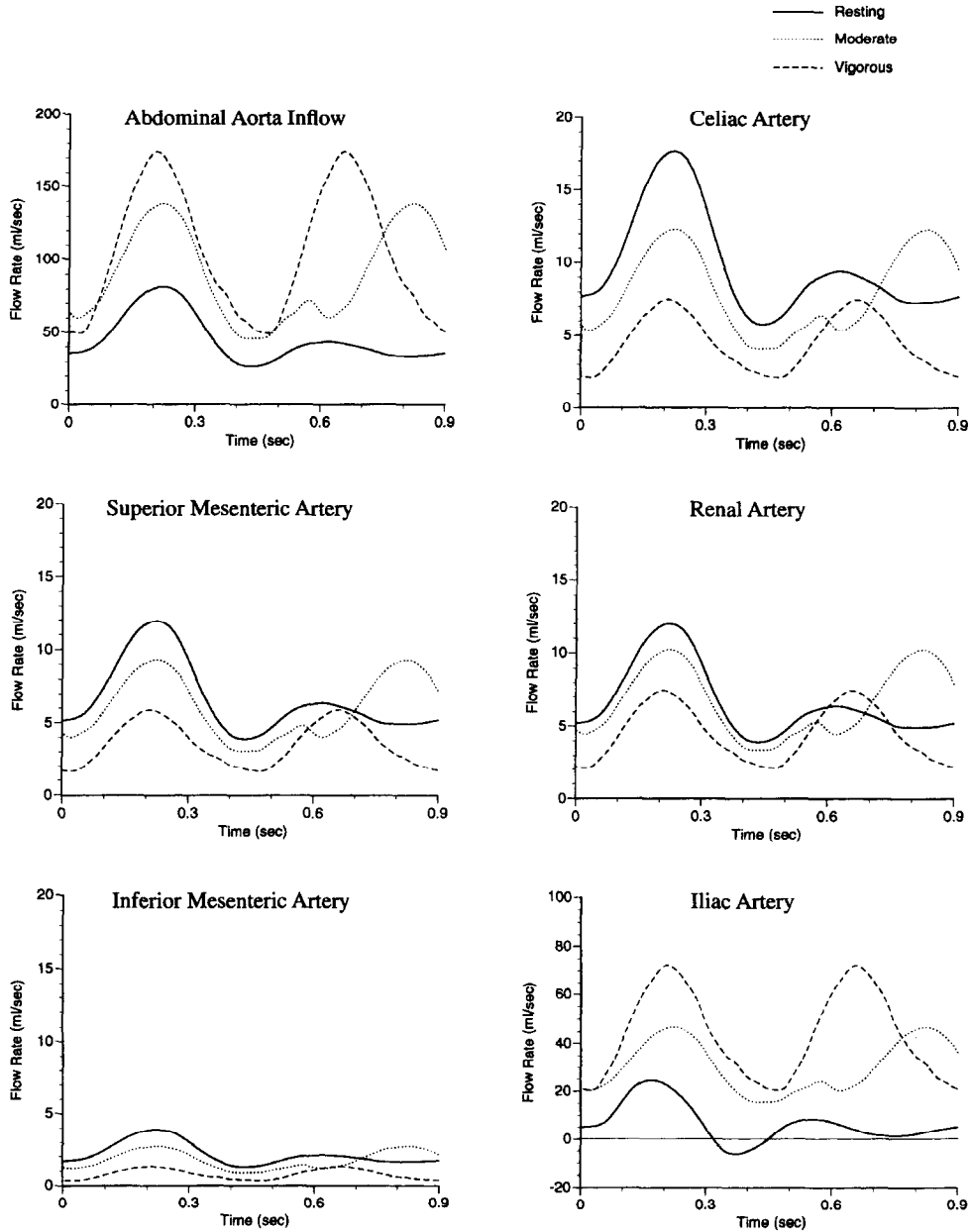


Fig. 42. Flow rate time functions for resting, moderate, and vigorous exercise conditions.

information age, it is probable that the use of computer models to understand vascular blood flow, vessel deformation, mass transport, and vascular remodeling will increase dramatically. As the models of vascular anatomy and physiology increase in complexity, the importance of having a comprehensive framework for computational vascular research will be of even greater import. The need for a comprehensive framework for computational vascular research motivated the construction of the Virtual Vascular Laboratory. Tasks that required weeks, such as the construction and modification of the model of the abdominal aorta, can now be performed in minutes. The geometric representation employed enables the use of an automatic mesh generator resulting in a high quality discretization with literally only seconds of user interaction. The knowledge-based environment used to create the Virtual Vascular Laboratory enables the complete specification of boundary conditions and solution strategies at the push of a button. The finite element solver employed results in time accurate solutions to hundreds of thousands of equations and permits the resolution of flow features heretofore

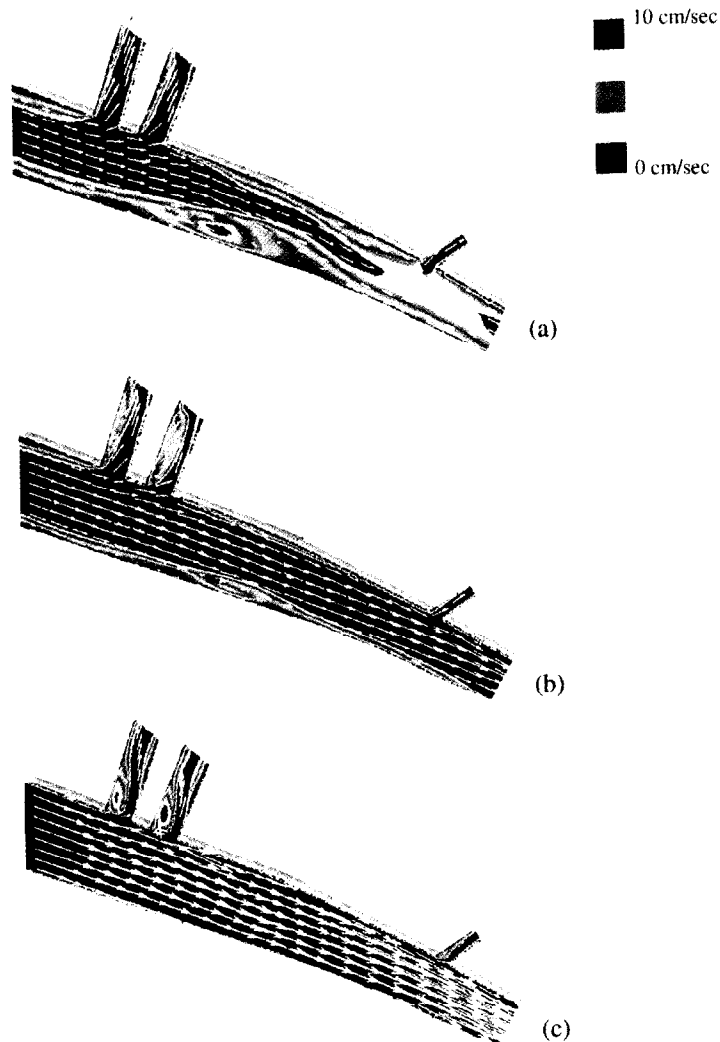


Fig. 43. Midplane slice of the abdominal aorta model displaying contours of axial velocity and the velocity vector field at mid-diastole for: (a) resting conditions; (b) moderate exercise conditions; (c) vigorous exercise conditions.

unobserved and unmeasurable. The scientific visualization techniques utilized allow the examination of vast amounts of data from multiple perspectives. Vector fields, streamlines, particles, and isosurfaces provide multiple perspectives of the velocity field. Contour maps and graph probes facilitate the qualitative and quantitative extraction of information on wall shear. Each component technology employed plays a key role in the successful solution of complex vascular hemodynamic problems.

Validation studies were performed to examine the spatial and temporal accuracy of the finite element method employed to solve the governing mathematical equations. Good agreement between the computational solution and the analytical Womersley solution was obtained for pulsatile flow in a long, straight artery. Very good agreement between the computational solution and experimental LDA data was obtained for steady and pulsatile flow in an end to side graft-artery anastomosis. A computational solution of this problem had not been previously reported.

The pulsatile flow in a model of an abdominal aorta was the most extensively studied problem in this paper. There have been no reported computational solutions and little quantitative experimental data for this important vascular blood flow problem. This example also illustrates the importance of the approach described in this paper. The geometry is far more complex than that of a single bifurcation and provided evidence of the efficacy of the knowledge-based object modeling, geometric solid modeling, and automatic mesh generation strategies

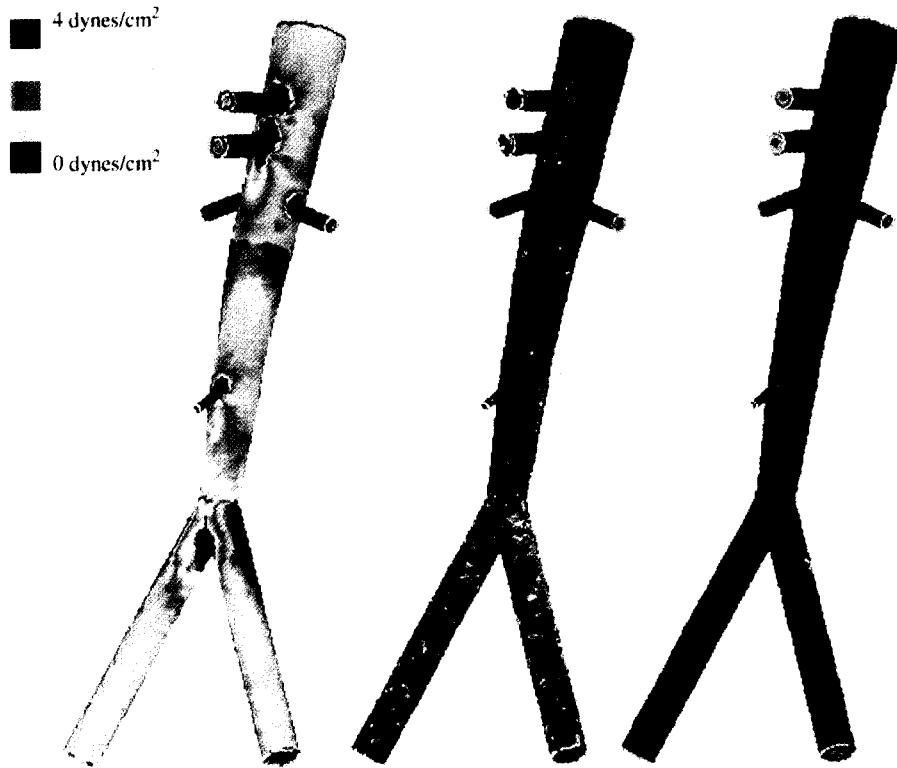


Fig. 44. Mean shear stresses for the abdominal aorta model under: (a) resting conditions; (b) moderate exercise conditions; (c) vigorous exercise conditions.

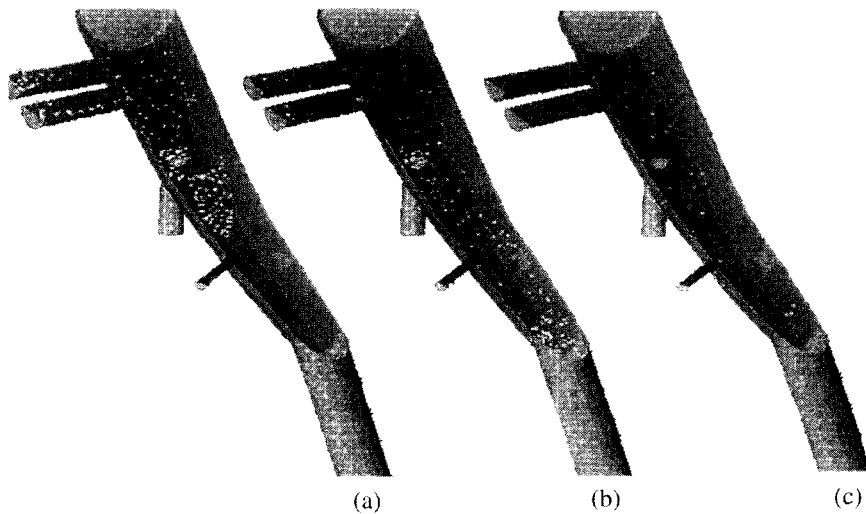


Fig. 45. Particles one second after release in the abdominal aorta model under: (a) resting conditions; (b) moderate exercise conditions; (c) vigorous exercise conditions.

employed. The size of the problem required the application of parallel finite element analysis software such as the SpectrumTM Solver used in the present approach. Multiple visualization techniques were utilized to extract information and obtain insight into the hemodynamic conditions in the abdominal aorta.

The clinical conclusions of the abdominal aorta study are believed to be equally significant. The complete temporal and spatial variations of the velocity field may enable new insights into particle residence, shear stress,

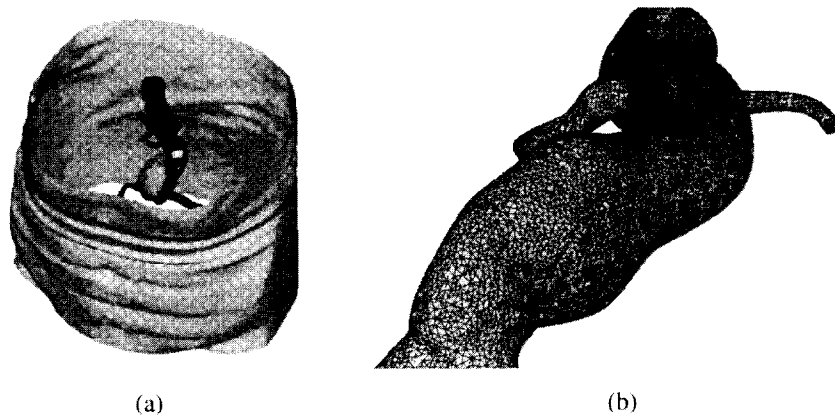


Fig. 46. 3-D reconstruction of an abdominal aorta from spiral CT. (a) Geometric model showing skin, blood flow domain, large clot, and calcifications; (b) close-up of finite element mesh in the aneurysmal region.

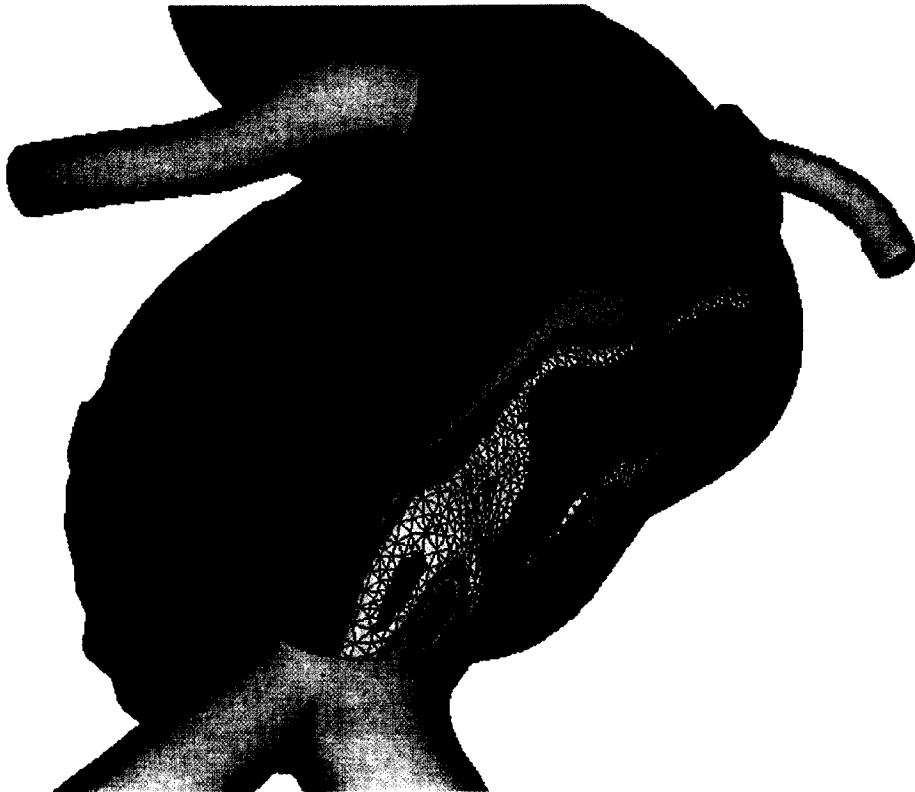


Fig. 47. Contour slice of velocity magnitude in the plane through the aortic bifurcation at end-systole.

and shear stress temporal and spatial gradients, and will help resolve the question of which hemodynamic factors correlate with the disease localization patterns observed in cadavers and in vivo. The information obtained regarding the effects of graded exercise on wall shear stress and particle residence times yields new insights into the effects of exercise and supports the growing body of evidence that moderate levels of exercise are beneficial in reducing vascular disease. These studies provide impetus to examine exercise hemodynamics in vivo using magnetic resonance imaging techniques, and to assess the validity of the assumptions made regarding the distribution of blood flow.

The final problem presented, the hemodynamic conditions in a patient-specific model of an aneurysmal

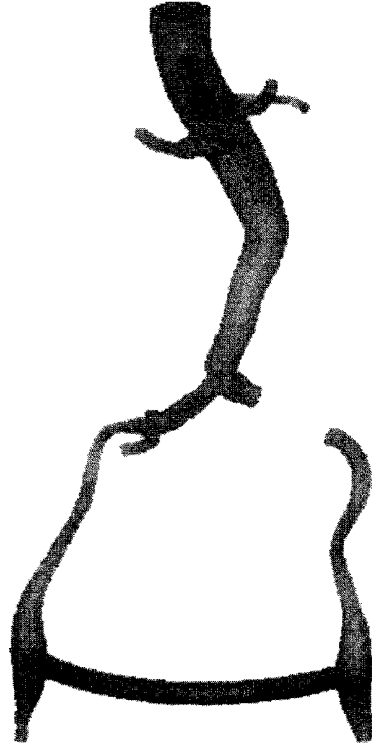


Fig. 48. Post-operative model of the patient-specific abdominal aorta.

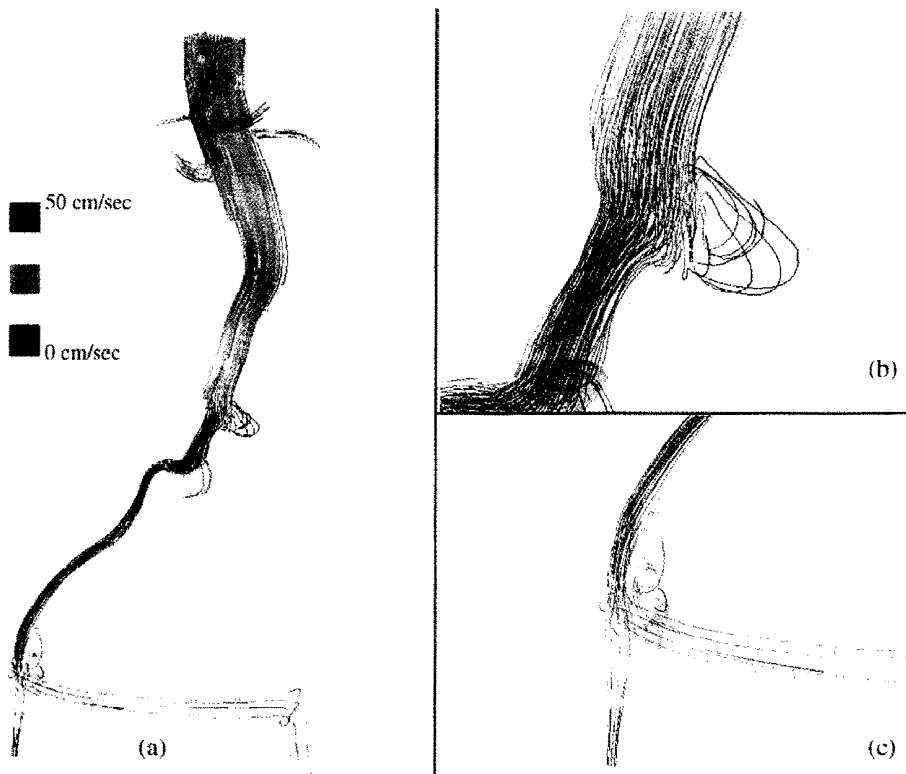


Fig. 49. Post-operative flow streamlines. (a) In entire model; (b) at aortic bifurcation; (c) at right femoral-bypass junction.

abdominal aorta, demonstrates for the first time the possibility of introducing predictive computational methods into vascular surgery planning. The approach implemented was essential for the timely construction of this vascular model from imaging data and the subsequent modification to reflect a possible surgical procedure. Computer aided surgical planning would not be possible for vascular surgery if it took weeks or months to create computational models, but it is possible if these models can be created in minutes or hours.

The key conclusion to be drawn from the software developments described in this paper is that in order to solve clinically relevant problems in vascular hemodynamics, the time invested early on to develop a comprehensive systems approach to the problem will pay off by minimizing the time required for creating new problems and solving them. Research in vascular hemodynamics could benefit from the integrated modeling, analysis, and visualization approach described herein.

There are numerous areas in computational vascular modeling which need to be addressed ranging from improved computational speed, further validation of solutions and assumptions, and new approaches to outflow boundary conditions, to the incorporation of models of artery wall mechanics and wall transport. In addition, the clinical application of predictive techniques is a research area that will occupy investigators for many years to come. Presumably, with knowledge of the effect of hemodynamic conditions on vascular adaptation and disease, and the knowledge of the actual hemodynamic conditions in an individual patient, better treatments can be devised to improve individual patient care.

Acknowledgments

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